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Improving Lifecycle Visibility of Household Appliances:

A QR-Based Approach to Sustainable Decision-Making in Property Management

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Abstract

This study investigates how information management and traceability of household appliances within property management can be improved to support more sustainable and informed decision-making. In many real estate organizations, limited access to structured lifecycle information regarding appliances, such as age, condition, maintenance history, and warranty status, leads to uncertainty in maintenance and renovation processes. As a result, functional appliances may be replaced prematurely, leading to unnecessary costs and environmental impact.

The study was conducted as a qualitative case study in collaboration with the company Fixkod, a technology company specializing in digital traceability solutions for property management. Empirical data were collected through semi-structured and open interviews with stakeholders within the real estate and related sectors, including operational management and technology-oriented roles. To further explore potential solutions to the identified findings, a QR-based traceability prototype was developed using an Artificial Intelligence (AI) - supported development tool. The prototype was iteratively refined based on empirical findings and evaluated through follow-up feedback interviews.

The findings indicate that current asset-related information is often fragmented, difficult to access, and insufficiently integrated into existing workflows. This contributes to reactive and risk-minimizing decision-making, limited forecasting capabilities, and challenges in supporting circular asset management practices such as repair and reuse. Organizational barriers, including resistance to change, system fragmentation, and operational access limitations, further complicate lifecycle management processes. The developed prototype illustrates how improved traceability and centralized information management could support more informed decision-making by increasing visibility of appliance lifecycle data, maintenance history, reuse status, and forecasting information. The study further highlights how digital traceability systems may support circular resource utilization by improving coordination of reuse activities, enabling more proactive maintenance planning, and reducing uncertainty in operational processes.

The study contributes to research within Supply Chain Management (SCM) by extending existing knowledge on traceability, visibility, and information management into an internal property management context characterized by distributed and access-constrained assets. Furthermore, it demonstrates how digital lifecycle tracking systems may support sustainable asset management and long-term resource efficiency within the real estate sector.

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1. Introduction

1.1 Background

Property management involves managing large amounts of physical resources, including home appliances. Within the built environment, resource use is nearly half of global resource extraction, one third of generated waste, and up to 12% of total national material-related emissions (Elnour et al., 2025). In Sweden, municipalities are responsible for their own waste management, while property owners bear the associated costs, including collection, transportation, treatment, recycling, and administration (Naturvårdsverket, 2025). Effective management of these resources is therefore critical, not only from a cost-efficiency perspective but also in relation to sustainable development.

However, many organizations lack structured information regarding the lifecycle of household appliances, such as age, condition, and maintenance history. This lack of information creates uncertainty during renovation processes, where appliances are sometimes replaced despite still being functional. Previous research has highlighted that replacement initiatives driven by sustainability goals often focus on improving energy efficiency (Alejandre et al., 2022). Such initiatives are not always based on comprehensive lifecycle data and may result in the premature replacement of functional products. This overlooks the environmental impact associated with manufacturing new appliances, including emissions from production and supply chains. Extending product lifecycles through maintenance and repair can significantly reduce environmental impact by delaying end-of-life and avoiding new production (Alejandre et al., 2022). A challenge within the construction and property management sector is the limited digitalization of asset-related information. Critical data required for maintenance and operations are often stored in fragmented or paper-based systems, leading to inefficiencies and communication challenges (Olimat et al., 2023). Despite the recognized importance of data for effective asset management, facility managers continue to face difficulties in accessing and utilizing relevant information.

From a SCM perspective, this issue can be understood as a lack of traceability and information flow within the organization's internal supply chain. In this context, the internal supply chain refers to the flow of household appliances and related information within the property management organization, including procurement, installation, maintenance, reuse, replacement, and disposal decisions. Appliances move between suppliers, apartments, maintenance processes, storage facilities, and reuse hub, while information about their location, condition, service history, warranty status, and lifecycle status needs to move between tenants, technicians, operational managers, and digital systems. While traceability and digital information systems are well-established areas of research within SCM, limited attention has been given to how these concepts can support lifecycle management of distributed household appliances within property management contexts. In practice, when

household appliances are procured and installed in residential units, information regarding their location, condition, and lifecycle status may become fragmented or difficult to access over time (Tsay et al., 2022). As a result, property managers may lack complete visibility regarding which appliances are installed where, their remaining useful life, and the maintenance history. This lack of traceability contributes to inefficient decision-making (Tsay et al., 2022), particularly during renovation projects, where appliances are frequently discarded regardless of their actual condition.

Additionally, these assets are located within residential units, which further complicates access to information. Physical inspection of appliances is often limited due to restricted access to tenants' living spaces, making it difficult to assess condition and status in a timely and efficient manner. This increases the reliance on accurate, structured, and accessible digital information for effective decision-making. In this study, a QR-based traceability prototype is explored as one possible response to the identified problem of fragmented appliance information, limited traceability, and decision uncertainty. A Quick Response code (QR code) is a machine-readable code that can be scanned using a smartphone or similar device to access digital information. The prototype is not studied as a technological solution in isolation, but as a way to examine how improved information visibility may support lifecycle management, reuse, and more sustainable decision-making within property management.

1.2 Purpose and Research Questions

The purpose of this study is to analyze current challenges related to information management and traceability of household appliances within property management, and to develop and evaluate a QR-based system aimed at improving sustainable resource utilization within internal supply chains.

To fulfill this purpose, the study addresses the following research questions:

RQ1: How is information about household appliances currently managed in property management, and what challenges arise due to limited traceability and information accessibility?

RQ2: How can a QR-based system be designed to improve information management and visibility?

RQ3: How can such a system support decision-making and contribute to more sustainable resource utilization?

1.3 Thesis Outline

The remainder of this thesis is structured as follows.
Chapter 2 presents the theoretical framework and relevant literature related to SCM, traceability, asset management, decision-making, sustainability, and digital information systems.
Chapter 3 describes the research methodology, including the research design, data collection, thematic analysis, and prototype development process.
Chapter 4 presents the empirical findings from the conducted interviews.
Chapter 5 introduces the developed prototype and presents the evaluation and feedback collected during the follow-up interviews.
Chapter 6 provides the analysis and discussion of the findings in relation to the theoretical framework and research questions.
Finally, *Chapter 7* presents the conclusions of the study, followed by contributions, limitations, and suggestions for future research.

2. Theoretical Framework

This chapter first presents the property management context and the operational challenges related to appliance information management. Thereafter, relevant theoretical concepts related to visibility, traceability, asset management, information systems, sustainability, and organizational factors are introduced in order to support the analysis of the study.

A literature review was conducted to identify relevant research related to asset management, circular economy, supply chain visibility, and digital solutions within property management. Scientific articles were collected through academic databases using keywords connected to the research topic.

The literature search included keywords such as "*Supply chain visibility", "traceability", "asset management",* and *"lifecycle management".* These were also combined with more specific terms including "*data visibility", "appliances", "circular economy" "private sector",* and *"repair vs replacement".* Boolean operators such as AND and OR were used to refine the searches, for example, "*traceability AND asset management"* and *"public sector AND internal supply chains",* to identify relevant academic literature. Additional keywords included "*Decision making"," risk", "adaptive systems", "decision theory",* and *"uncertainty".*

The selected literature formed the theoretical foundation of the study and supported the analysis of challenges related to information visibility, lifecycle management, decision-making, sustainability, and organizational adoption within property management contexts.

2.1 Property Management Context

Information accessibility has been identified as one of the most common challenges in facility management. During the operational phase, facility managers often lack quick and efficient access to the information required to process maintenance tasks and work orders. This limits their ability to make informed and timely decisions (Tsay et al., 2022; Olimat et al., 2023). A central issue is referred to as the asset information gap, which originates from traditional methods of transferring asset-related information. Historically, asset information has been delivered in physical documents, binders, or scattered digital files, which are rarely maintained in a structured or accessible manner over time. As a result, available information is often described as inadequate, inefficient, and costly to manage (Tsay et al., 2022)

Several specific challenges related to asset information have been identified by Tsay et al. (2022). First, a large proportion of data is still manually entered into systems, a process that is both time-consuming and error prone. Second, digital records are frequently incomplete. Previous studies have shown that a significant share of assets lack essential information such as maintenance procedures, serial numbers, and warranty details. Third, even when data exists, it is often stored in ways that make it difficult to retrieve. In such situations, the effort required to locate and interpret information may exceed the cost of replacing the asset, thereby reducing the practical value of the information itself.

Organizations often struggle to define their information requirements, resulting in data that is not aligned with operational needs. In addition, asset data collection during construction phases is frequently deprioritized, leading to incomplete datasets (Tsay et al., 2022). These challenges illustrate the difficulties associated with managing distributed assets within property management, where limited accessibility and fragmented information reduce the effectiveness of both decision-making and lifecycle management.

2.2 Supply Chain Visibility and Information Sharing

SCM emphasizes the importance of effective coordination of material and information flows. A central concept within SCM is visibility, which refers to the extent to which relevant information is accessible across different actors and processes. Traditionally, organizations have viewed information as a source of power, resulting in reluctance to share data. However, previous research suggests the opposite, namely that increased information sharing enhances overall system performance by reducing uncertainty and improving responsiveness (Christopher & Lee, 2004). Information-rich supply chains support improved coordination, faster decision-making, and increased operational efficiency. From an internal supply chain perspective, visibility is equally important, as limited access to information may result in inefficiencies, fragmented processes, and suboptimal decision-making. Within property management contexts, limited visibility may reduce an organization's ability to coordinate maintenance activities, monitor asset status, and support lifecycle-oriented decision-making.

In this study, the internal supply chain refers to the flows of appliances, information, maintenance activities, reuse processes, and decision-making within the property management organization. Household appliances move through internal processes such as procurement, installation, use, maintenance, storage, reuse, and replacement. At the same time, information about these appliances must flow between tenants, technicians, operational managers, and internal systems. From this perspective, fragmented appliance information can be understood as a disruption in the internal supply chain, limiting visibility and reducing the organizations' ability to coordinate resources effectively.

2.3 Traceability in Supply Chains

Traceability refers to the ability to track and trace the history, location, and status of products throughout their lifecycle. Within supply chains, traceability enhances transparency and enables improved monitoring of operations. It contributes to improved risk awareness, improved reliability, and enhanced trust among stakeholders (Razak et al., 2023). Furthermore, traceability supports the early detection of disruptions and enables organizations to respond more proactively. By facilitating real-time tracking and information exchange, traceability strengthens supply chain resilience and flexibility, allowing resources to be redeployed efficiently when disruptions occur (Razak et al., 2023). Despite these benefits, traceability remains limited in practice, particularly in contexts such as residential property management, where assets are geographically distributed and often difficult to access physically. Traceability can be supported through different digital technologies, including barcodes, Radio Frequency Identification (RFID), Near Field Communication (NFC), Internet of Things (IoT), and integrated asset management systems. These technologies differ in terms of cost, complexity, automation level, and implementation requirements (Al-Maktary & Susanto, 2025). For example, RFID and IoT solutions may enable more automated tracking, while simpler code-based solutions can provide a low-cost link between physical assets and digital information records.

2.4 Asset Management and Lifecycle Perspective

Asset management focuses on the systematic management of physical assets throughout their lifecycle, including acquisition, operation, maintenance, and disposal. A lifecycle perspective emphasizes that decisions regarding assets should be based on long-term value rather than short-term considerations (Diop et al., 2021). Effective asset management requires accurate data on asset condition, usage, and maintenance history. During the early phases of the lifecycle, such as acquisition and installation, decisions are made that influence long-term operability, maintainability, and sustainability (Diop et al., 2021). Without reliable lifecycle information, organizations risk inefficient maintenance strategies and premature replacement

of assets. This highlights the importance of structured lifecycle information when managing household appliances within distributed property portfolios.

2.5 Information Systems and Decision-Making

Information systems play a critical role in supporting decision-making processes within organizations. Effective decision-making depends on access to accurate, timely, and relevant information. In modern organizations, the volume of available data has increased significantly, making structured information management essential for maintaining operational efficiency and competitiveness. Inadequate information systems may result in fragmented data, poor communication, and reduced decision quality. As highlighted by Berisha-Shaqiri (2014), the lack of well-structured information systems limits an organization's ability to utilize information effectively, which negatively impacts operational performance. This challenge is particularly evident in asset-intensive environments, where information is often fragmented, dispersed across multiple systems, or poorly managed. Within property management contexts, fragmented information systems may therefore contribute to reduced visibility and inefficient lifecycle management practices.

2.6 Sustainability and Circular Resource Use

Sustainability within operations and SCM emphasizes the efficient use of resources and the reduction of waste. Circular economy principles aim to extend product lifecycles through repair, reuse, and maintenance, thereby reducing environmental impact. Previous research indicates that sustainability-driven replacement initiatives often focus primarily on energy efficiency while neglecting a broader lifecycle perspective (Alejandre et al., 2022). As a result, functional products may be prematurely replaced despite the environmental impact associated with manufacturing new appliances. Extending product lifecycles through maintenance and repair can significantly reduce environmental impact by delaying end-of-life and avoiding new production. Additionally, previous studies show that although there is a willingness among users to repair appliances, replacement remains common due to convenience and limited access to information (Delemere & Liston, 2024). This highlights the need for improved systems that support informed decision-making regarding repair, reuse and replacement decisions.

2.7 Organizational Factors in Decision-Making

This section presents organizational and decision-making factors that are relevant for understanding both current asset management practices and the potential adoption of a new

way of working. Decisions regarding repair, reuse, replacement, and procurement are not only influenced by available information, but also by existing routines, perceived risks, user acceptance, and organizational willingness to change. Therefore, public and private sector decision-making, uncertainty, and resistance to change are discussed as important perspectives for analyzing how transparency systems may be adopted and used in practice.

2.7.1 Public vs. Private Sector

Previous research challenges the common assumption that public sector organizations are inherently more risk-averse and resistant to change compared to the private sector organizations. While public organizations are often perceived as cautious, empirical studies suggest that this distinction is more nuanced (Nicholson-Crotty et al., 2019). For instance, Nicholson-Crotty et al. (2019) demonstrate that public managers are not consistently more risk-averse than their private sector counterparts. Their findings indicate that decision-making is highly context-dependent and influenced by how outcomes are framed. When decisions are presented in terms of potential losses, public sector managers may even demonstrate higher levels of risk tolerance. In addition, factors such as public service motivation and altruistic goals can encourage risk-taking when decisions are perceived to benefit society.

These findings suggest that resistance to change in the public sector cannot be fully explained by individual risk aversion alone. Instead, such behavior may be more closely related to organizational structures, institutional constraints, and contextual factors influencing decision-making.

From an innovation perspective, entrepreneurship has been described as a driver of change in both public and private sectors. While some emphasize entrepreneurship as a disruptive force, others frame it as a process of managing change through negotiation and coalition-building. Although originating from different theoretical perspectives, both approaches highlight the importance of individuals in initiating change rather than responding to existing demand (Pedersen, 2022).

Furthermore, public sector organizations are often subject to higher levels of external scrutiny and reputational pressure. According to Lee et al. (2023), public organizations frequently face negative stereotypes regarding inefficiency, regardless of actual performance. As a result, maintaining legitimacy becomes critical, which may contribute to more cautious and justifiable decision-making, particularly in situations involving uncertainty. This suggests that decision-making behavior is shaped not only by individual preferences but also by organizational context and external pressures.

2.7.2 Decision-Making under Uncertainty

These contextual factors become particularly relevant in property management because decisions regarding repair, reuse, replacement, and procurement are often made under uncertainty. Uncertainty is an unavoidable factor in decision-making, particularly in situations involving risk, long-term consequences, or complex alternatives. It often arises when outcomes are difficult to predict, especially when decisions may result in losses or suboptimal outcomes (Schultz et al., 2010). Uncertainty can be understood as a lack of knowledge and is commonly categorized into input uncertainty and model uncertainty. Input uncertainty relates to incomplete knowledge about relevant variables, while model uncertainty concerns the assumptions and relationships used to describe a process (Schultz et al., 2010). Although some uncertainty can be reduced by collecting additional information, this is often costly, time-consuming, or practically difficult.

In contexts where information regarding assets is limited or difficult to access, decision-making may rely on assumptions, experience, or simplified rules rather than structured data. This may influence decisions regarding repair, replacement, or reuse. Schultz et al. (2010) further emphasize that decision analysis should be introduced early in the process, as the framing of a decision influences what data is collected and how uncertainties are addressed. They also highlight the importance of iterative approaches, where analysis is refined as new information becomes available.

From this perspective, improved information visibility and traceability can reduce uncertainty by enabling more informed, structured, and data-driven decision-making.

2.7.3 Resistance to Change

In addition to decision-making challenges, organizational factors also influence whether new systems can be adopted and used effectively in practice. Resistance to new information systems is often driven not by the technology itself, but by the organizational and human-related changes that implementation entails (Seo, 2013). One important driver is the perceived loss of control. Increased transparency and standardized processes may reduce individual autonomy, particularly in environments where employees are accustomed to managing their own workflows independently. Similarly, concerns about job security may arise when systems automate tasks or alter existing roles.

In smaller organizations, resistance may also stem from individuals who hold critical tacit knowledge. In such contexts, standardized systems may be perceived as a threat to individual roles and organizational importance (Coşkun et al., 2022). The transition to new systems is often associated with increased workload, particularly during the implementation phase, which may contribute to user reluctance (Seo, 2013). Resistance may also manifest through workarounds, such as the use of “shadow systems” (e.g., spreadsheets), or through

inconsistent data entry, both which may reduce overall system effectiveness (Haug et al., 2009; Seo, 2013).

The level of resistance may vary depending on organizational characteristics. Smaller organizations often experience stronger resistance due to reliance on informal processes, while larger organizations are typically more accustomed to standardized systems and formalized workflows (Coşkun et al., 2022; Haug et al., 2009). To mitigate resistance, previous research highlights the importance of structured change management. This includes strong top management support, targeted training initiatives, and clear communication aimed at increasing user acceptance and reducing uncertainty during implementation processes (De Búrca et al., 2005). Resistance to change is therefore relevant not only as a general implementation issue, but as a factor that may determine whether improved traceability actually leads to better information management, or whether users continue relying on existing routines, informal communication, or parallel systems.

2.8 Research Gap

In summary, previous research highlights the importance of visibility, traceability, information systems, lifecycle management and organizational factors in achieving efficient and sustainable supply chain operations. However, these concepts have primarily been studied within traditional industrial and inter-organizational supply chain contexts.

Limited research has explored how these concepts can be applied within internal supply chains in property management, particularly in relation to the lifecycle tracking of household appliances located in residential environments. In such contexts, assets are geographically distributed and often physically inaccessible, which further complicates information management and decision-making processes. In addition, organizational factors such as existing routines, perceived risks, user acceptance, and integration with current workflows may influence whether digital traceability systems can be adopted and used effectively.

This creates a gap in understanding how digital traceability systems can support maintenance, reuse and replacement decisions in distributed and access-constrained environments. Consequently, there is a need to further explore how improved information visibility and traceability can enhance both operational efficiency and sustainable resource utilization within property management. This study aims to address this gap by exploring how a QR-based traceability system can support lifecycle management, information accessibility, and decision-making in a property management context.

3. Methodology

This chapter presents the methodological approach used in the study. It describes the research design, data collection methods, participant selection, thematic analysis process, and the development of the QR-based prototype. In addition, the chapter discusses ethical considerations, credibility, and limitations related to the chosen methodology.

3.1 Research Design

Given the constrained ten-week timeframe, this study adopted a qualitative dominant research strategy aimed at exploring how information management and traceability of household appliances could be improved within property management, and how digital solutions could support decision-making and sustainability. The study was conducted as a case study within the Swedish property management sector, allowing for an in-depth investigation of current practices and challenges related to asset management and information flow. Semi-structured interviews served as the primary method of data collection. Interviews were conducted with relevant stakeholders, such as Operational Managers, technicians, project managers, and tenants, in order to gain insights into how information about household appliances was currently managed, what challenges exist, and how decisions regarding maintenance and replacement were made.

In addition, a prototype of a QR-based traceability system was developed using Lovable, an AI-supported tool which turns descriptions into code and develops the system. The prototype functioned as an artifact intended to illustrate a potential solution and was used during follow-up interviews to gather feedback on usability, functionality, and perceived value. This enabled an evaluation of how digital traceability could improve information accessibility and decision-making. A total of six interviews were conducted during the study, each lasting approximately forty minutes. Purposeful sampling was applied to select participants with relevant knowledge, professional experience, and insight into the research topic. Access to participants was facilitated through collaboration with the technology company Fixkod.

Although the study was primarily conducted as a qualitative case study, the research process also shares similarities with a design science research logic. The study identified a practical problem within property management, developed an artifact in the form of a QR-based traceability prototype, evaluated the artifact through feedback interviews, and refined it based on the identified needs and responses. However, the study does not claim to follow a full design science methodology but rather uses a design-oriented and iterative logic to support exploration of how digital traceability may address the identified information management challenges.

3.2 Literature Review

The literature review supported the development of the theoretical framework, the interview guide, and the analysis of the empirical findings. Relevant literature related to asset management, traceability, supply chain visibility, information systems, and sustainability was reviewed in order to identify key concepts and previously identified challenges connected to the research topic.

3.3 Data Collection

Empirical data were collected through semi-structured interviews. This method was chosen to allow flexibility during the interviews while still ensuring that key topics relevant to the research questions were covered. The interview guide was structured around themes such as current asset management practices, information handling, decision-making processes, and challenges related to appliance management (see Appendices A, B & C). A total of six interviews were conducted with participants selected based on their relevance to the research topic and their professional experience within the field. Purposeful sampling was applied to ensure that the participants possessed relevant domain-specific knowledge and experience. The respondents represented different perspectives, including property management, different hierarchical perspectives in the real estate sector, and digital solution development, thereby contributing to a broader understanding of the studied context.

In addition to the semi-structured interviews, open feedback interviews were conducted during the later stages of the study in connection with the developed prototype. These interviews aimed to gather participant reflections regarding usability, functionality, and practical applicability of the proposed QR-based traceability system. The open interview format allowed respondents to freely discuss ideas, concerns, and suggest improvements based on the presented prototype, which also contributed to the iterative refinement of the system design.

The interviews were conducted both digitally and in person and were recorded with the participants' consent. Each interview was subsequently transcribed to enable systematic analysis of the empirical material.

3.4 Data Analysis

To analyze the collected data, thematic analysis was applied using the six-step framework proposed by Braun & Clarke (2006). This method was selected due to its suitability for identifying patterns and themes within qualitative data.

The analysis followed an iterative process. Initially, the transcripts were read multiple times to achieve familiarity with the empirical data. Subsequently, initial codes were generated based on relevant segments of the data. These codes were then continuously refined and compared across interviews in order to identify similarities, differences, and recurring patterns. In the following step, the codes were grouped into broader themes and sub-themes, which were visualized using mind map diagrams created in Canva, a visual material design tool (see Appendices D, E, F & G). Throughout the analysis process, the themes were continuously reviewed and refined to ensure internal consistency and clear distinctions between categories. The final themes reflect recurring patterns identified within the empirical data and form the basis for the empirical findings presented in the study.

3.5 Prototype Development

As part of the study, an exploratory prototype of a QR-based traceability system was developed. The purpose of the prototype was not to create a fully functional system, but rather to illustrate a potential digital solution and support the evaluation of how improved traceability could enhance information accessibility, lifecycle management, and decision-making. The prototype allows each appliance to be assigned a unique QR code, which can be scanned to access and update information such as installation date, condition, lifecycle status and maintenance history. A QR-based approach was selected because the main challenge identified in the study concerned access to structured lifecycle information rather than real-time automated tracking. QR codes provide a simple and low-cost way to connect a physical appliance to a digital information record using devices that are already widely available, such as smartphones.

The prototype was developed using Lovable and was iteratively refined based on insights from both the empirical findings and the follow-up feedback interviews. It was subsequently used during open feedback interviews to gather participants' reflections regarding usability, functionality, and perceived value within a property management context.

The development process can be understood as consisting of three main iterative stages. In the first stage, insights from the literature review and initial empirical findings were translated into core design requirements, such as the need for appliance-level traceability, lifecycle visibility, reporting functionality, and reuse support. In the second stage, an initial prototype was developed and presented during follow-up feedback interviews. Respondent feedback concerned the prototype's functionality, usability, practical relevance and perceived value. In the third stage, the prototype was refined based on this feedback by adjusting interface elements, adding or clarifying functions, and strengthening the connection between lifecycle data, operational decision-making, and sustainability indicators. This iterative process allowed the prototype to develop from a general traceability concept into a more context-specific proof of concept for property management.

3.6 Research Quality

To ensure the credibility of the findings, the analysis was conducted systematically and transparently following an established methodological framework. The iterative coding process, together with continuous comparison across interviews, contributed to strengthening the reliability of the identified themes.

Due to the time-constrained nature of the thesis, the study was based on six interviews. This limits the generalizability of the findings, and the results should therefore be understood as context-specific insights rather than universally applicable conclusions. However, the study provides a foundation for future research to further explore the QR-based traceability and lifecycle-oriented asset management within property management contexts.

Despite the limited number of interviews, the selected participants provided relevant professional experience and domain-specific knowledge that contributed to a meaningful understanding of the research problem. In addition, the inclusion of participants from different organizational roles and perspectives contributed to increased credibility through methodological triangulation and broader contextual understanding.

3.7 Sustainable and Ethical Aspects

This study considers ethical, societal, and sustainability-related aspects of digital traceability within property management. By adopting a broader perspective, the study aims to move beyond purely technical and operational implications and instead examine the wider consequences of improved information management within internal supply chains.

Ethical considerations in this study primarily relate to data handling, transparency, and responsible use of information. Since the study involves interviews with employees from a real estate company and the use of organizational information, it is essential to ensure informed consent, voluntary participation, and confidentiality. Furthermore, ethical responsibility also includes ensuring that the proposed QR-based system supports fair and transparent decision-making regarding asset replacement, without leading to misuse of data or unintended negative consequences for stakeholders. In addition, the use of QR-based systems raises considerations related to data security and access control. Since QR codes provide direct access to asset-related information, it is important to ensure that only authorized users are able to retrieve and modify data. Without appropriate safeguards, there is a risk that sensitive information may be accessed, altered or misused. Therefore, the design of such systems should consider fundamental cybersecurity principles, including controlled access, secure data handling, and user authentication. Although these aspects are not the primary focus of this study, they remain important considerations for future implementation of digital traceability solutions.

Societal aspects relate to how improved traceability, and digitalization may influence decision-making processes within property management. Enhanced access to information about appliances may contribute to more informed and transparent decisions, which may affect not only organizational efficiency but also tenant experience and trust. In addition, the introduction of digital systems may influence work practices for property managers and technicians, requiring adaptation to new ways of handling information and maintenance processes.

Sustainability considerations are central to this study, as the proposed system aims to improve resource utilization and reduce unnecessary waste. By enabling improved visibility of appliance lifecycle data, such as age, condition, and maintenance history, the system has the potential to extend product lifetimes and reduce premature replacement. This may contribute to reduced *environmental* impact through lower material consumption, decreased waste generation, and improved alignment with circular economy principles.

By integrating ethical, societal, and sustainability dimensions, this study aims to provide a holistic understanding of how digital traceability systems may support more responsible, efficient, and sustainable decision-making within property management and internal supply chains.

3.8 Use of AI

AI tools such as ChatGPT were used solely for language refinement and grammar support during the writing process. No AI-generated content was used as a substitute for the author's own analysis, interpretations, or academic reasoning. All theoretical discussions, empirical analysis, and conclusions reflect the author's own work and understanding and have been critically reviewed and edited accordingly. AI-supported tools were also utilized during the development of the prototype as part of the design process.

4. Empirical Findings

This chapter presents the empirical findings from the conducted interviews. The findings are structured according to the main themes identified through the thematic analysis process inspired by Braun and Clarke's (2006) six-step framework. The purpose of this chapter is to describe how household appliance information is currently managed, what challenges respondents identified, and what factors influence decisions related to maintenance, reuse, replacement, and lifecycle management.

The chapter is organized into four main themes. The first theme, Current Asset Management Practices, describes existing systems, maintenance processes, and reuse practices within property management. The second theme, Data Visibility and Asset Traceability, focuses on

how asset-related information is stored, accessed, and used, as well as the consequences of fragmented and inaccessible data. The third theme, Operational Circularity in Asset Management, presents findings related to repair, reuse, and barriers to circular asset management. The fourth theme, Operational Constraints and Limited Physical Access, addresses organizational resistance, physical access limitations, and decision-making under uncertainty.

Together, these themes provide an empirical foundation for understanding the current challenges in appliance lifecycle management and for later analyzing how improved traceability may support information accessibility, decision-making, and sustainable resource utilization.

4.1. Current Asset Management Practices

The findings describe how asset management is currently carried out within the studied organization, including existing systems, processes, and practices. While structured approaches are already in place, the findings also indicate how these structures influence the way information is collected, accessed, and utilized in practice. The findings within this theme are divided into three sub-themes: existing data storage and system structure, operational processes for asset management, and existing circular practices (see Appendix D). Due to the overlap between sub-themes, the findings are presented under one combined heading for clarity.

Respondents described current asset management practices as being structured around centralized property management systems and reactive maintenance processes. Asset-related data is stored within property management systems, including information such as apartment allocation, service history, and warranty related information. According to the respondents, this provides a certain level of visibility and allows employees to access relevant information when needed.

"[...] then we take it out of the system, install it, and then the system is updated with the fresh data." – Operational Manager, Large Real Estate.

Maintenance processes are primarily initiated through customer-reported faults, after which technicians are dispatched to assess the issue. Decisions regarding repair or replacement are typically based on technician evaluations and communicated back to operational management for further decision-making.

"*As fast as the tenant reports that something is wrong with their fridge for example, then that is turned into an errand that is stored on the apartment [...] There is always a technician that is sent out to check if it can be fixed or not, then they report to us, and we take the decision to change." – Operational Manager, Large Real Estate.*

In addition, respondents described the use of internal reuse practices, where functional appliances are collected, stored, and reused across different apartments when possible.

4.2. Data Visibility and Asset Traceability

The findings indicate that while asset-related data is stored within existing systems, the information is not always structured, integrated or easily accessible, which affects how effectively it can be used. Challenges related to fragmented data and limited visibility impact both decision-making and planning. The findings are divided into three sub-themes: importance of data management, consequences of limited data visibility, and fragmented and inaccessible asset data (see Appendix E).

4.2.1 Importance of Data Management

Respondents described data management as a central component of asset handling in property management. Access to structured and reliable information was considered important for understanding which assets are installed, where they are located, and what maintenance or replacement activities have previously been performed. Several respondents emphasized the importance of maintaining a comprehensive overview of appliances, including service history, warranty information, and documentation connected to specific units.

"[...] If there were no money restriction, no restriction in resources like employees and so on, I would like a system that, if you go into an apartment number, every single appliance in that apartment would appear, where it is and when it is installed, warranty. Basically, I could just easily go in and get access to it all." – Operational Manager, Large Real Estate

Existing systems were described as useful when information is registered consistently, for example during installation or when maintenance activities are performed. In such cases, asset-related data may provide a clearer overview of appliance status and historical activities. Respondents also highlighted the value of systems that function as centralized records of information, as these enable relevant actors to retrieve information when needed.

"Yes, when you install a new appliance, it is also registered in the real estate system." – Operational Manager, Large Real Estate

Overall, the findings indicate that data management is regarded as important not only for operational purposes but also for creating traceability, continuity, and improved visibility within long-term asset handling processes.

4.2.2 Consequences of Limited Data Visibility

Several respondents described limited visibility of asset-related information as a factor that affects both planning and decision-making processes. When there is no clear overview of

appliance history, age, or condition, organizations may experience difficulties in assessing whether appliances should be repaired, replaced, or reused. This was particularly emphasized in larger renovation projects, where multiple apartments and assets are managed simultaneously.

"[…] What is missing is an easy way to register and gather the information that is needed at that second. Some real estate systems do have support for this, although what is missing is an easy way to access this information." *– Stefan Hvalgren, Fixkod*

Limited data visibility was also described as negatively affecting forecasting and long-term planning. Without an accessible overview of installed assets and their lifecycle status, it becomes harder to anticipate future replacement needs or estimate procurement volumes over time. Respondents explained that this may create uncertainty and inefficiencies in operational planning processes.

"[…] If we for example, only buy in half, and it then later shows that we need to change all of them. Then the building will be delayed, and there will be much bigger consequences. Therefore, it is much easier to round up the amount." – Stefan Hvalgren, Fixkod

In practice, lack of visibility was also described as contributing to unnecessary procurement or premature replacement, particularly in situations where existing assets cannot easily be assessed, traced or identified. According to the respondents, this creates challenges not only from a financial perspective, but also in relation to resource utilization and long-term planning.

4.2.3 Fragmented and Inaccessible Asset Data

Although asset-related data often exists within organizations, respondents described it as fragmented and difficult to access in practice. Information may be stored across multiple systems, connected to different actors, or remain dependent on individual knowledge rather than being available in a centralized and unified format. As a result, important asset-related information is not always easily retrievable when needed.

"If a technician is out on a job, and changes parts or alike, it is hard to register it. Since they do not share the same IT system. The information is probably shared through mail or similar, where they say what they have done. But what serial number did the washer have?" – Stefan Hvalgren, Fixkod

Respondents also described challenges related to appliance identification and continuity of information replacement processes.

"[…] Today you could have for example, a laundry room with 10 floor washing machines. When we receive for a fault report, which appliance is it for? You need to have an identifier as a color code or a number. But if the washing machine is changed, then this number or color code does not go with it. Then you don't have anything, so when the tenants do send in a fault

report, then it is hard to know which machine it is that needs maintenance." – Operations manager, Large Real Estate.

Physical accessibility constraints were also highlighted as contributing to these challenges. Since many appliances are located inside tenants' homes, obtaining updated information regarding a specific unit may require arranging physical access to the apartment. According to the respondents, this creates a higher threshold for inspection and limits the possibility of verifying information directly on site.

In addition, fragmented information was described as complicating the organization of a comprehensive overview of the asset base. Key information such as appliance age, maintenance history, or lifecycle status may be unclear or difficult to compile, particularly across larger property portfolios. These findings, therefore, indicate that the challenge is not necessarily the total absence of data, but rather the lack of accessible, consolidated, and usable information.

"*There hasn't been a good system to keep track of which appliances are above their lifecycle and which ones that needs to get replaced." – Stefan Hvalgren, Fixkod*

4.3 Operational Circularity in Asset Management

The findings reveal several challenges that limit the ability to fully utilize the lifecycle of appliances. While respondents described existing practices that support reuse and circular handling of appliances, limitations related to information accessibility and decision-making processes were found to affect the effectiveness of circular asset management. The findings within this theme are divided into three sub-themes: reactive and case-based decision-making, barriers to circular asset management, and enablers of circular asset management (see Appendix F).

4.3.1 Reactive and Case-Based Decision-Making

Respondents described that decisions related to appliances are typically made on a case-by-case basis. Rather than following standardized procedures or relying on structured lifecycle data, decisions are often based on the specific condition and circumstances of each individual asset. Factors such as cost, perceived condition, and input from technicians were described as central in determining whether an appliance should be repaired or replaced.

"Everything is a matter of cost […] If the refrigerator's compressor has stopped working and is maybe around 8-10 years old. Well then, you change the whole fridge because it is not feasible to keep and repair. It is a decision that is taken case-by-case." – Operational Manager, Large Real Estate

In many situations, physical inspections are required before a decision can be made. Technicians are dispatched to assess the condition of the appliance, after which recommendations are communicated back to the operational management. This indicates that decision-making is strongly dependent on on-site evaluations rather than being driven by centralized, predictive data or lifecycle-based information.

"It could be that the silicone strip or sealing strip needs to be changed. But there will always be a person who looks at it and sees if it can be fixed or not, then he sends it back to us, and we decide to change." – Operational Manager, Large Real Estate

Several respondents also described how cost considerations influence these decisions. If an appliance is perceived as too old or too costly to repair, replacement is often viewed as the more viable alternative. The findings therefore suggest that decision-making processes are largely reactive and dependent on immediate assessments rather than long-term planning or lifecycle optimization.

4.3.2 Barriers to Circular Asset Management

Respondents highlighted several challenges that limit the ability to fully utilize the lifecycle of appliances. One commonly described issue was that functional assets may be replaced even though they still have remaining lifespan, particularly in larger renovation contexts where multiple apartments and assets are handled simultaneously.

"The big problem is that the appliances are located inside tenants' homes, which makes it difficult to run and check things […] We had multiple discussions with people in management organizations within the real estate sector. They said it is difficult and tedious, and that it is much easier just to pay for new appliances than to fulfil the whole lifecycle of the appliance." – Stefan Hvalgren, Fixkod

In addition, respondents described the lack of clear and accessible information regarding asset history and condition as a significant barrier to reuse. When information about appliance age, maintenance history or previous usage is unclear or difficult to access, it becomes more challenging to assess whether an appliance is suitable for continued use. According to the respondents, this may contribute to more cautious decision-making, where replacement is prioritized over repair or reuse.

Respondents also emphasized that planning and coordinating reuse activities can become particularly complex in larger organizations and property portfolios. Without a clear overview of available appliances, their condition, and their location, coordinating reuse across different apartments or storage facilities becomes difficult. This may result in situations where functional appliances are not utilized efficiently despite still having remaining functional value.

4.3.3 Enablers of Circular Practices

Despite these challenges, respondents described several existing practices that support circular asset management. One commonly mentioned example was the use of internal reuse systems, where functional appliances are collected, stored and later reused in other apartments. According to the respondents, this enables organizations to extend the lifecycle of appliances and reduce unnecessary waste.

"[...] when we are doing a total renovation of the estate, we make sure to take care of fridges, freezers, and ovens in our own reuse hub. Were we put in the appliances from time to time, that can later be used in another apartment, which means that they get a reused oven, for example." – Operational Manager, Large Real Estate

Respondents also emphasized that improved control over asset-related information could support more sustainable practices and resource-efficient practices. Access to structured information regarding appliance condition, maintenance, history, and usage was described as important for enabling more informed decisions related to repair, replacement, and reuse.

In addition, respondents described repair as an important strategy for extending appliance lifecycles and reducing unnecessary replacement. Maintaining appliances through spare-part replacement was viewed as preferable when feasible, particularly for appliances that still retained functional value.

"*We try to repair, of course, everything. Spare parts should exist up to 10 years for fridges or freezers. Usually, it can be plastic packings and alike that breaks, then we try to buy new parts before they need to be changed." – Operational Manager, Large Real Estate*

Overall, the findings indicate that although circularity is not always systematically implemented, several existing practices and attitudes within the organizations already support more resource-efficient and lifecycle-oriented asset management.

4.4 Operational Constraints and Limited Physical Access

The findings highlight several constraints that affect asset management practices, particularly in relation to physical accessibility and organizational conditions. These limitations influence both how information is obtained and how operational decisions are made. The findings within this theme are divided into three sub-themes: organizational and cultural resistance to change, physical and operational access constraints, and decision-making under constraints and uncertainty (see Appendix G).

4.4.1 Organizational and Cultural Resistance to Change

Respondents described the property management sector as relatively conservative, characterized by established routines and a generally limited willingness to change existing ways of working. Several respondents highlighted that both employees and organizations often prefer to continue with familiar processes rather than adopting new systems or operational practices.

"*[…] Most organizations don't want to change, and there are some sectors that are more static than others […] where they have done it in one way for many years, real estate is the same thing, renting out an apartment is not something new […] it's a very static sector." – Stefan Hvalberg, Fixkod.*

This resistance was described as existing on both an individual and an organizational level. Users may be reluctant to adopt new tools if they perceive them as unnecessary, disruptive, or difficult to integrate into existing workflows. At the organizational level, respondents explained that stability, predictability, and risk avoidance are often prioritized over experimentation and innovation. In particular, public-sector organizations were described as demonstrating slower adoption of new solutions due to perceived risks and limited incentives for organizational change.

"[…] It's a lot of the feeling of control, the governance part. That you know your organization is just as important." – Stefan Hvalberg, Fixkod

At the same time, respondents emphasized that successful implementation of new systems often depend on internal support and engagement. The presence of motivated individuals, internal ambassadors, or dedicated teams was described as important for driving change and facilitating adoption across the organization.

"One obstacle is that the realtors find other solutions. They could be worse, but they solve it in another way and are not willing to change. And secondly, they can see it as an obstacle to start with a new system if it cannot be integrated with their current system." – Hampus Granström, Project Leader Internationalization

"What is needed first is a person in the customer-organization that liked the idea, an ambassador. And then it is needed that the organization allows for changes." – Stefan Hvalgren, Fixkod

4.4.2 Physical and Operational Access Constraints

A central challenge described by respondents is that many assets are located inside tenants' homes, which significantly limits direct physical access. This creates practical constraints when attempting to inspect, verify, or collect information about specific appliances. According to the respondents, gaining access often requires coordination with tenants, increasing the effort and time required to assess asset conditions.

"[…] When it is inside a tenant's apartment then it is a much bigger threshold to gain access and get inside the tenant's home, it is nearly impossible." – Stefan Hvalgren, Fixkod

As a result, physical inspections are not always feasible on a larger scale, particularly in organizations managing a large number of geographically distributed assets. Respondents explained that this limits the ability to continuously maintain updated information regarding individual appliances and contributes to uncertainty within asset management processes.

In practice, this was described as creating reliance on indirect methods of obtaining information, primarily through technician visits initiated by tenant fault reports. This further reinforces a reactive maintenance approach, where information is updated mainly when issues arise rather than through continuous monitoring or preventive management practices.

"*[…] We have already procured, so it is easy to take quick decisions it is not needed to sit and make investigations, instead if the technician says that the compressor is dead and it is 10 years old, then we change the fridge." – Operations manager, Large Real Estate*

4.4.3 Decision-Making under Constraints and Uncertainty

Respondents described decision-making processes as often being conducted under conditions characterized by limited information and operational constraints. In situations where complete or easily accessible data is lacking, decisions are frequently based on experience, assumptions, or immediate assessments rather than structured and comprehensive information.

This was described as contributing to risk-minimizing behavior, where decisions are made in ways that reduce uncertainty rather than optimize long-term resource utilization. For example, replacing an appliance may be perceived as a safer and more predictable option than attempting repair when the condition, history, or remaining lifecycle of the asset is unclear.

Respondents also highlighted challenges related to forecasting and long-term planning. Limited access to consolidated and accessible information was described as making it difficult to predict future replacement needs, estimate procurement volumes, or plan maintenance activities proactively. According to the respondents, this contributes to reactive decision-making processes and may result in inefficiencies within both procurement and maintenance operations.

"*It is hard to know if a service has been provided and in what extent. If you clean thousands of staircases then you know, of course, that some are cleaned. But do you want to pay 100% of the bill and only 90%, 80% or 70% has been delivered?" – Stefan Hvalgren, Fixkod*

5. Prototype

This chapter presents the developed QR-based traceability prototype. The chapter describes the prototype development process, the system's core functionalities, and how the prototype was refined based on empirical findings and feedback from follow-up interviews. The purpose of the prototype is to illustrate how digital traceability solutions may support improved information accessibility, lifecycle management, decision-making, and sustainable resource utilization within property management.

To develop the prototype, the identified needs and findings from both the empirical data and the literature review were first collected and analyzed. These insights were then translated into design requirements, which formed the foundation for the prototype development process.

The prototype was created using an AI-supported development tool, where the identified requirements were formulated into a structured prompt describing both the intended functionality and the desired design of the system. The development process followed an iterative approach, meaning that the prototype was continuously revised and refined throughout the study based on emerging insights, identified limitations, and participant feedback.

The prompt focused on designing a modern, clean, and user-friendly system intended to support property managers in handling appliances throughout their lifecycle. The system was designed to improve visibility of asset-related information and support maintenance processes, decision-making, reuse practices, and sustainability-related operations.

An initial version of the prototype was presented during follow-up interviews, where respondents provided feedback regarding usability, functionality, and potential improvements. The final version of the prototype presented in this chapter, therefore, includes several revisions and refinements based on the feedback gathered during these interviews.

5.1 Prototype Features and Interface

The prototype includes several core functionalities, where the main dashboard page serves as a central entry point for users. Navigation is organized through a sidebar interface located on the left side of the system, providing access to the different system categories (see Figure 1). The interface is divided into the following sections: *Dashboard, Home appliances, laundry rooms, Reports, Technician, Sustainability, Reuse hub, Forecast* and *Tenant report*. Access to the different sections is role-based depending on the user type. For example, tenants only have access to the tenant reporting interface, while Operational Managers are provided access to all system functionalities. This structure was designed to simplify navigation and ensure that users only interact with information relevant to their role within the process.

The visual design of the prototype follows a clean and intuitive interface inspired by Scandinavian design principles, emphasizing simplicity, clarity, and ease of use. Features such as QR code integration, filtering functionalities, and visual status indicators were included to reduce complexity, improve visibility, and support faster decision-making processes.

It is important to note that all statistical values and displayed data within the prototype are fictional and were created solely to demonstrate the intended functionality and structure of the system. The prototype therefore does not contain real organizational data.

5.1.1 Dashboard

The purpose of this dashboard is to provide a comprehensive overview of the system, while simultaneously highlighting the sustainability impact of repair and reuse activities within the organization. The dashboard presents key performance indicators through visual elements, enabling users to quickly access important operational information. These indicators include the total number of tracked appliances, units requiring attention, open fault reports, and the number of reused appliances currently active in the system (see Figures 1, 2, 3 & 4). By presenting this information in a simplified and centralized format, the dashboard is intended to support quick decision-making and improve overall data visibility.

In addition, a circular diagram visualization is included to illustrate the lifecycle status of appliances across the system. This provides an intuitive overview of the condition of the asset base and enables users to identify how many appliances are functioning normally, require maintenance, or are approaching end-of-life (see Figure 1).

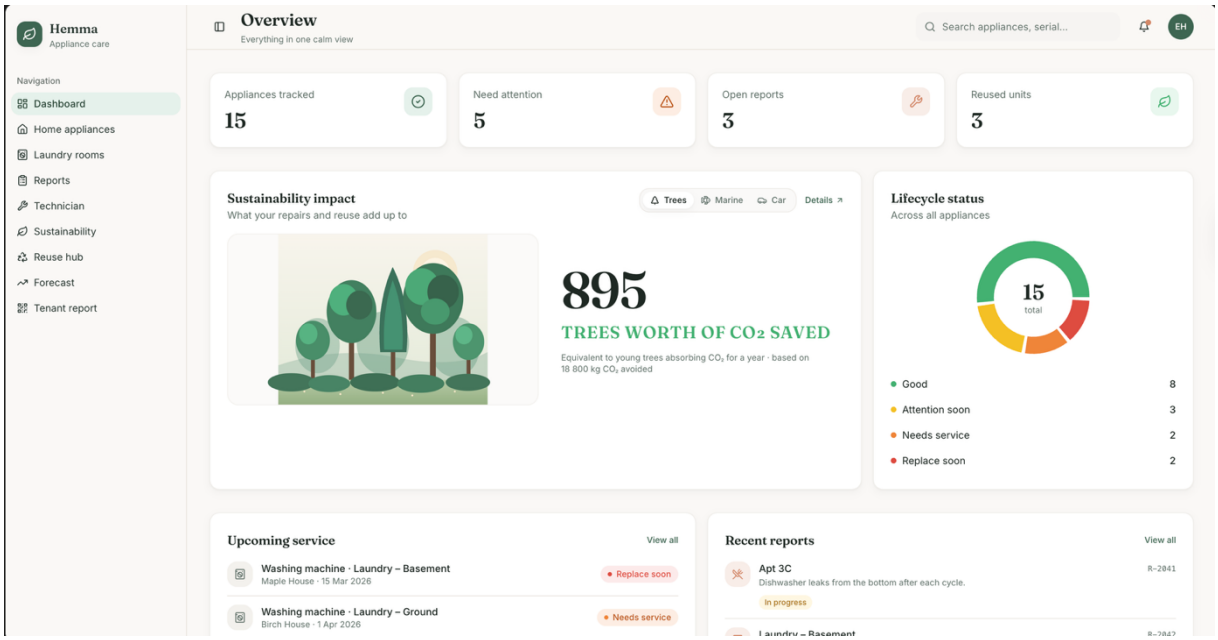


Figure 1 Dashboard overview with navigation menu and tree-based sustainability indicator.

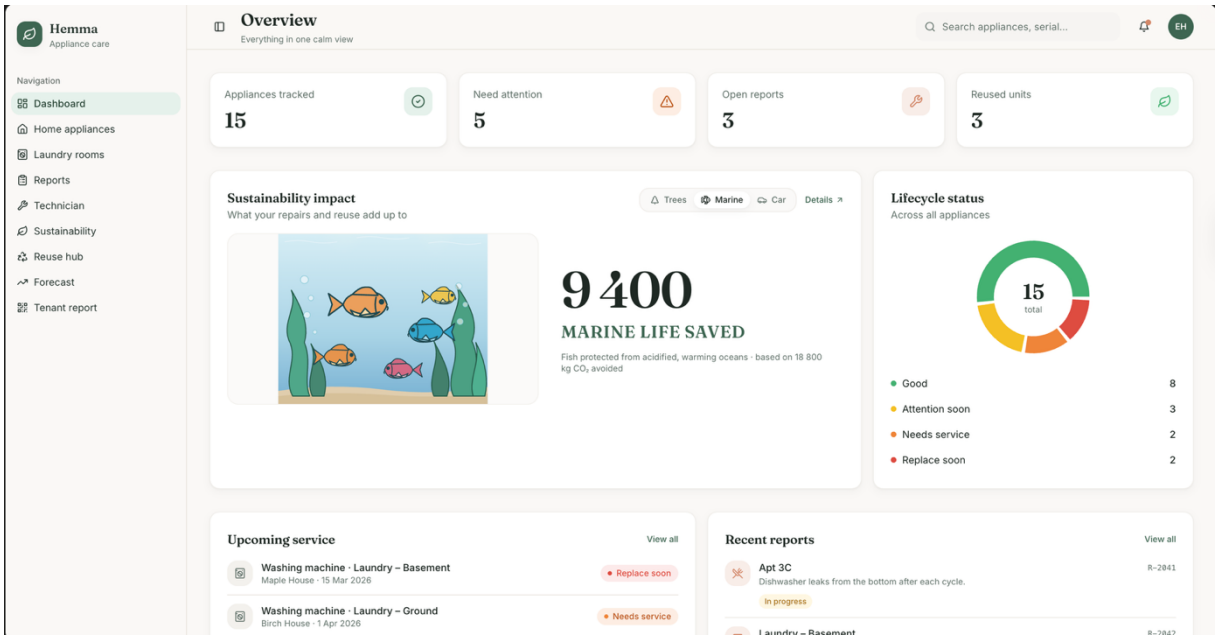


Figure 2 Dashboard view with marine life sustainability indicator.

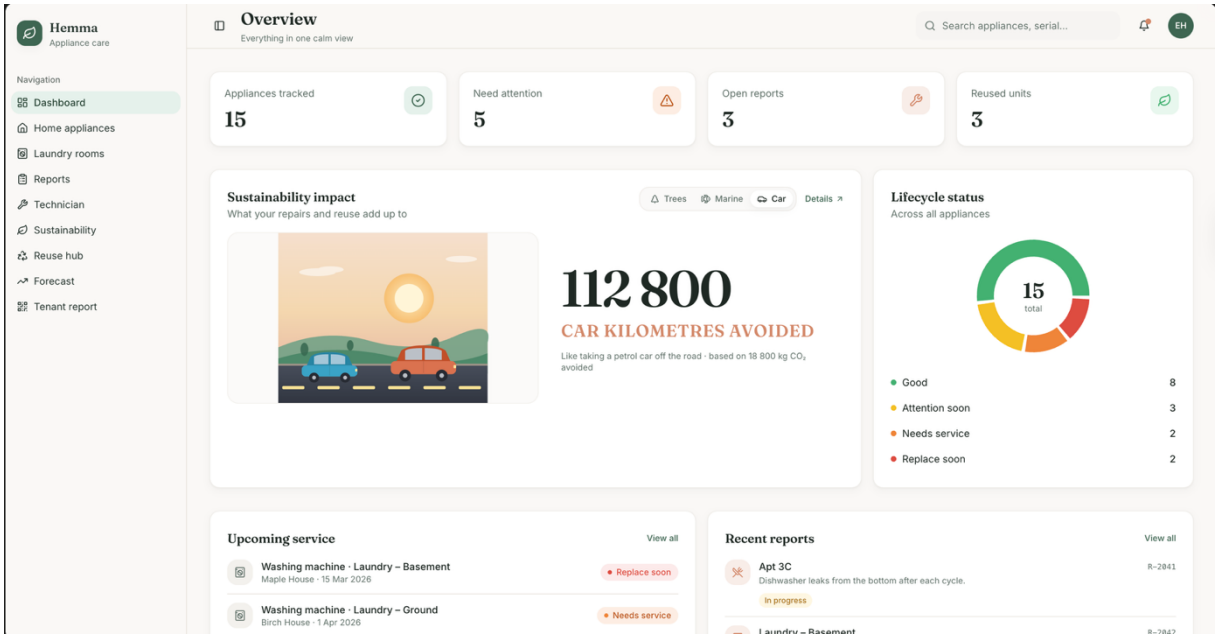


Figure 3 Dashboard view with vehicle-emission sustainability indicator.

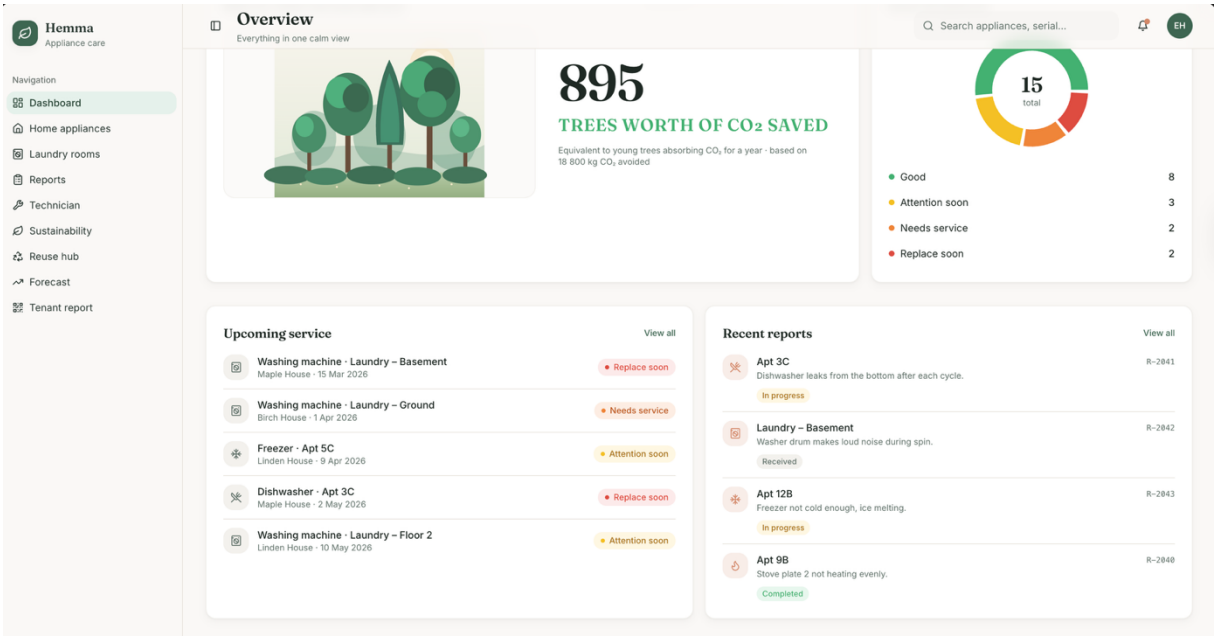


Figure 4 Lower section of the dashboard showing upcoming services and recent reports

To further support operational awareness, the dashboard also contains sections displaying upcoming service needs and recently reported issues (see Figure 4). These components are designed to improve accessibility to time-sensitive information and reduce the effort required to identify ongoing or upcoming operational tasks. The dashboard also includes simplified sustainability indicators intended to make environmental impact easier to understand and interpret. Rather than relying solely on graphs and numerical values, sustainability impact is additionally visualized through more intuitive and engaging indicators, such as trees saved, marine life protected, and reduced vehicle emissions (see Figure, 1, 2 & 3). The purpose of these visualizations is to create a clearer and more motivating representation of lifecycle management and sustainability performance. This approach aims to make sustainability related information more accessible and emotionally engaging for users, while still complementing the more detailed sustainability statistics available elsewhere within the system.

Overall, the dashboard is designed to improve visibility, support operational decision-making, and emphasize the sustainability benefits of extending appliance lifecycles through repair, reuse and maintenance activities.

5.1.2 Home Appliances and Laundry Rooms

To support the tracking and management of appliances, the system separates assets into two primary views: Home appliances and Shared laundry room. These views allow users to navigate between different operational contexts depending on where the appliances are located, thereby improving usability and clarity within larger property portfolios. Within these views, the system provides a detailed overview of each appliance, including information such as appliance type, serial number, location, installation date, lifecycle status, service history,

and upcoming maintenance needs. If an appliance is a reused appliance, it is visualized with a green leaf icon next to the name with the text “reused” (see Figure 5). This allows users to quickly identify reused assets while simultaneously gaining a broader overview of the asset base.

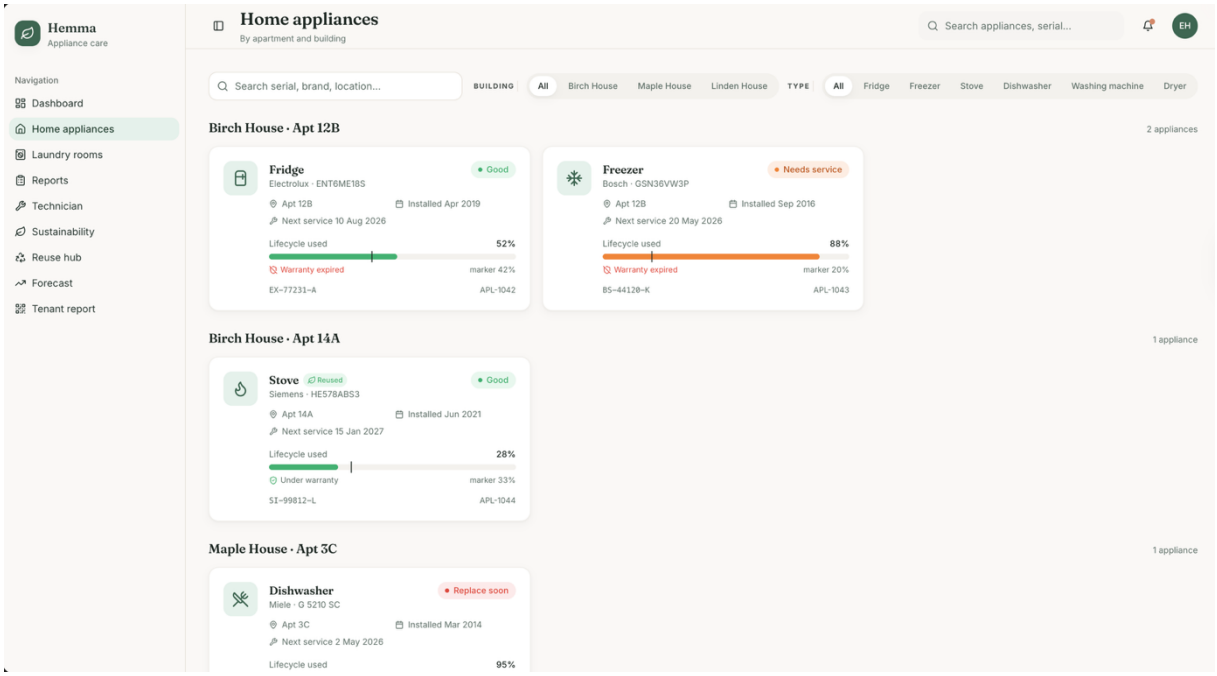


Figure 5 Home appliances overview with lifecycle and warranty indicators

To support lifecycle-based decision-making, each appliance is visualized using a lifecycle indicator presented by a progress bar displaying the percentage of the expected lifecycle already utilized, as well as the remaining lifespan (see Figure 5). In addition, a warranty indicator shows whether the warranty remains active or has expired (see Figure 5). Information regarding the upcoming service is also presented to support maintenance planning.

A search function is included to allow users to quickly locate specific appliances, while filtering options enable sorting based on factors such as location or appliance type. These features are intended to improve accessibility and reduce the time required to retrieve relevant information. Appliances are also visualized using a color-coded system reflecting their current condition and urgency level, providing users with an intuitive overview of which assets are functioning normally, require attention, or are approaching end-of-life.

The laundry room view for shared laundry rooms includes an additional functionality related to usage tracking. Within this interface, users can monitor how frequently laundry rooms appliances are used, providing insight into utilization level and demand. This information may support decisions related to resource allocation and appliance utilization, for example by identifying underutilized machines. If usage falls below a predefined threshold, the system

highlights this and suggests potential actions, such as relocation or removal of the unit (see Figure 6).

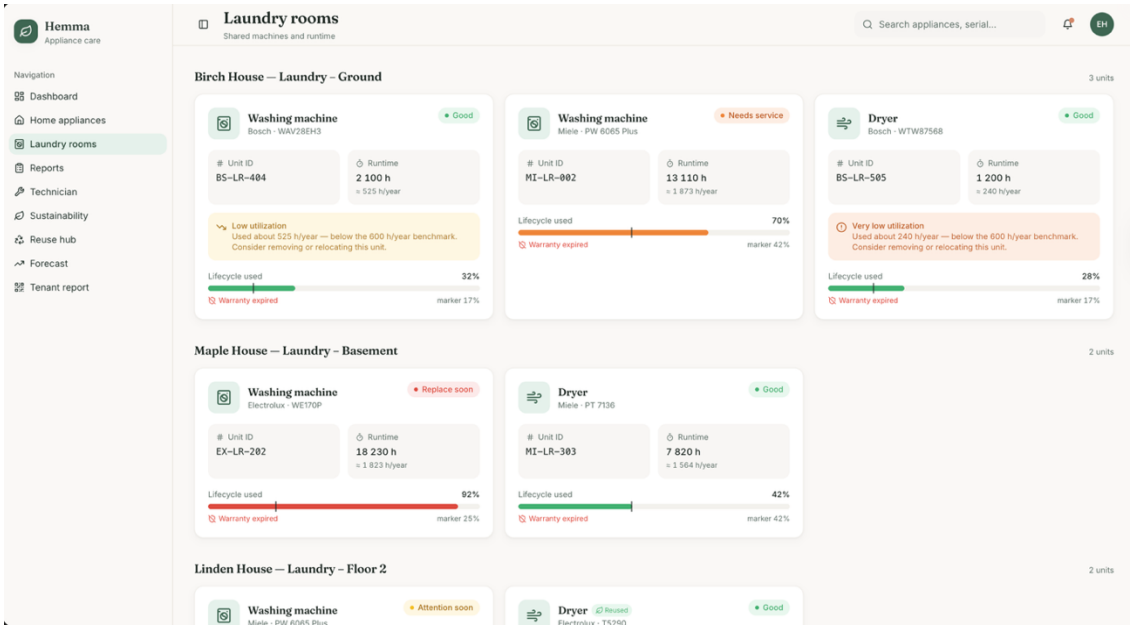


Figure 6 Laundry room appliance overview with usage tracking

Overall, these functionalities are intended to improve data visibility, support lifecycle-based decision-making, and enable more efficient management of distributed assets. In this way, the system addresses several of the challenges related to fragmented information and limited lifecycle visibility identified in the empirical findings.

5.1.3 Tenant Report

In addition, a simplified fault reporting interface was developed for tenants, enabling them to report appliance-related issues directly through the system. Reports can be submitted either by scanning a QR code attached to the appliance or by selecting the corresponding appliance registered within the apartment (see Figure 7). This dual functionality was designed to support accessibility and accommodate different user preferences and technical capabilities.

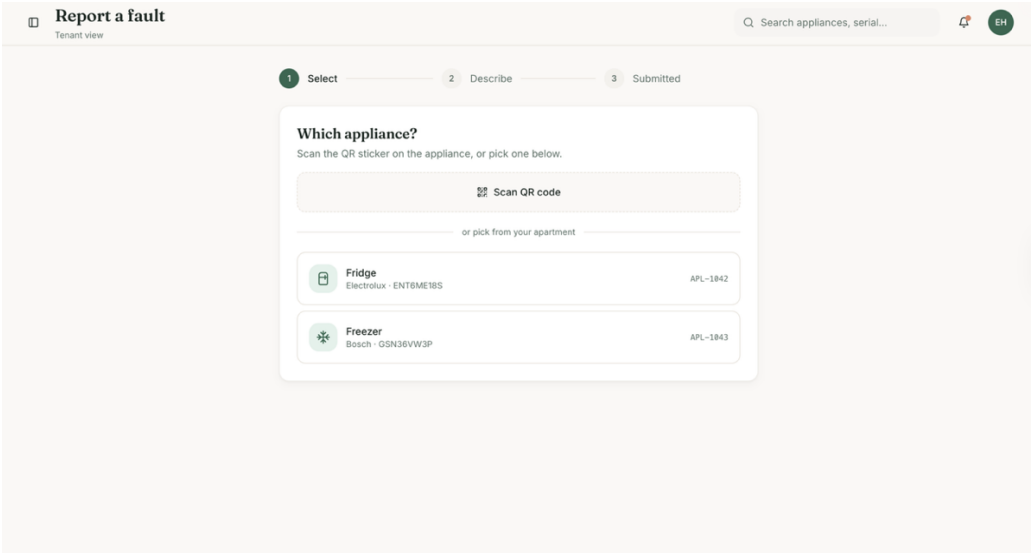


Figure 7 Tenant reporting interface for selecting an appliance

Once an appliance is selected, the tenant is directed to a reporting interface where the issue can be described, with the option to attach images if necessary (see Figure 8).

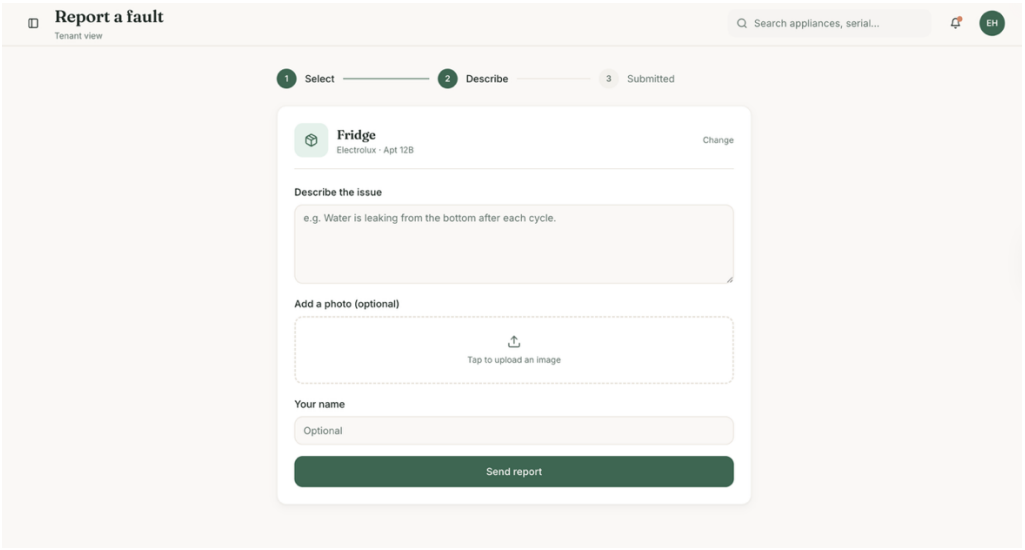


Figure 8 Tenant reporting form after appliance selection

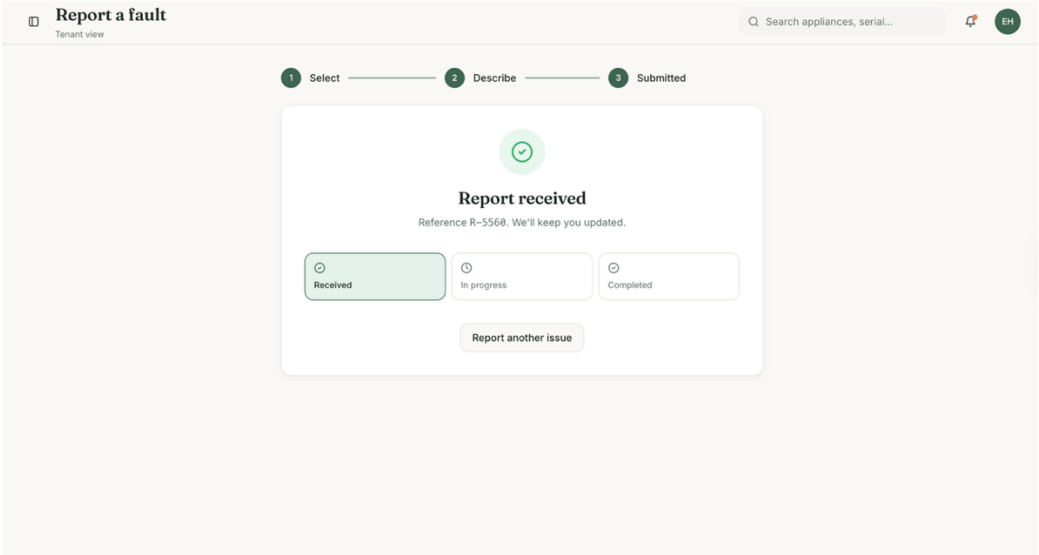


Figure 9 Tenant confirmation page showing report status

After submitting the report, the tenant receives a confirmation indicating that the issue has been successfully registered within the system. The system also allows tenants to track the progress of the report through status updates indicating whether the issue has been received, is currently in progress, or has been completed (see Figure 9). This functionality was designed to increase transparency and improve communication between tenants and property management.

Submitted reports are automatically integrated into the system and become visible within the Operational Manager's interface. From there, reports can be assessed and assigned to a technician for further action.

5.1.4 Reports

The reports page functions as a dedicated reporting interface for Operational Managers, where reported tasks, service updates, and system-wide information can be managed. The interface provides an overview of all incoming reports, which can be filtered into different categories such as received, in progress, and completed. This allows managers to quickly monitor the status of maintenance activities and gain a clearer overview of ongoing operations.

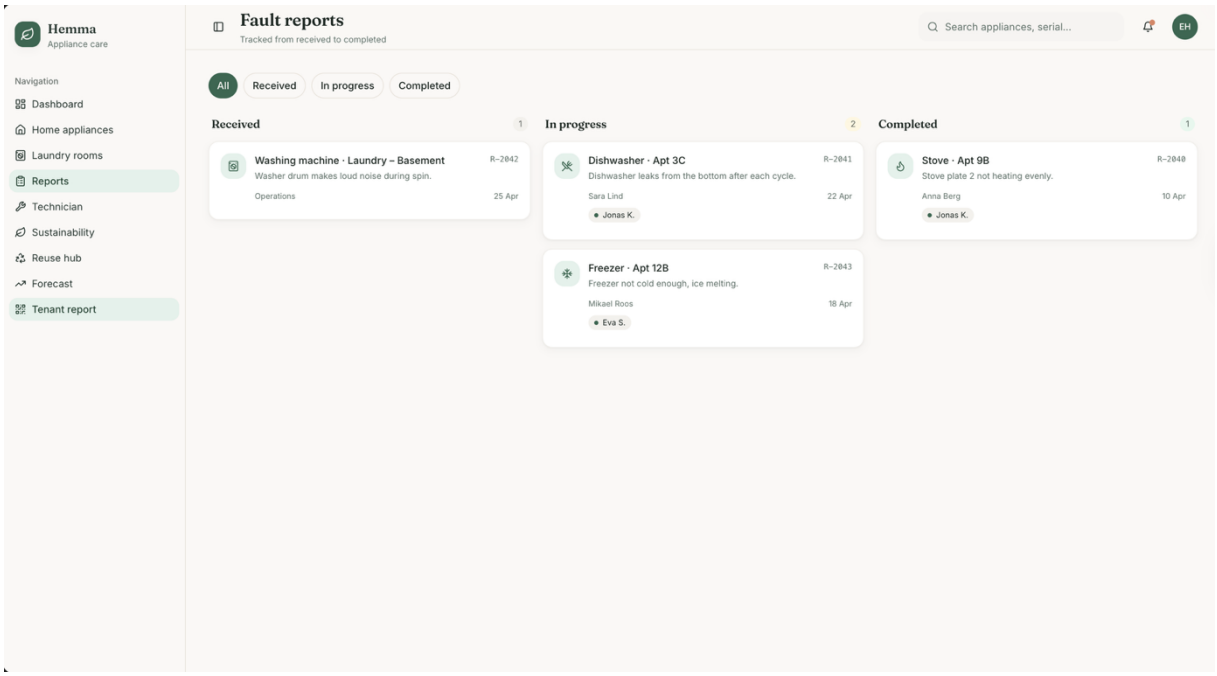


Figure 10 Reports overview

All reports are automatically registered within the system when a tenant submits a request, ensuring real-time visibility for the Operational Manager. As the maintenance process progresses, the status of each report is continuously updated by the assigned technician. For example, if a technician identifies the need for spare parts or is unable to complete a repair, this information is clearly highlighted within the system. Such cases are visually emphasized to reduce the risk of critical issues being overlooked and to support faster operational response (see Figure 11).

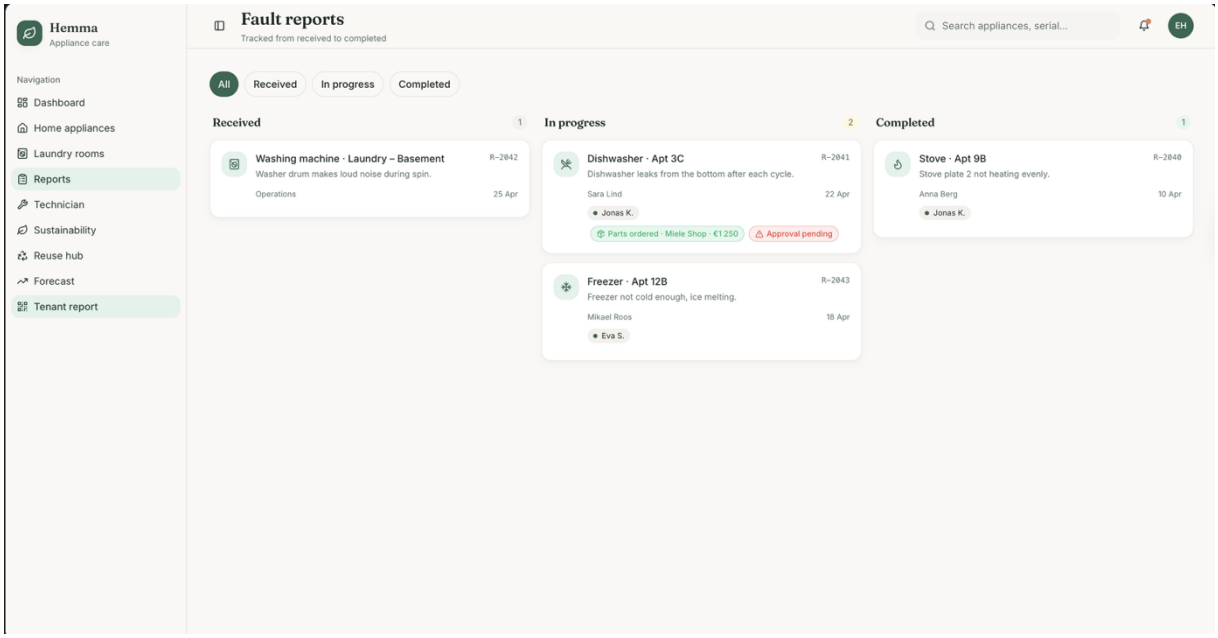


Figure 11 Reports interface showing unapproved spare-part order

The systems also include functionality for monitoring spare-part orders directly through the reporting interface. Users are able to see which parts have been ordered, their associated costs,

and whether the order exceeds predefined financial approval thresholds. If an order exceeds the predefined threshold without managerial approval being confirmed by the technician, the system highlights the order in red as a warning indicator (see Figure 11). Once managerial approval has been confirmed, the indicator changes to green, signaling that the purchase has been approved and can proceed (see Figure 12). This functionality is intended to maintain financial control while simultaneously allowing technicians to work more autonomously and efficiently.

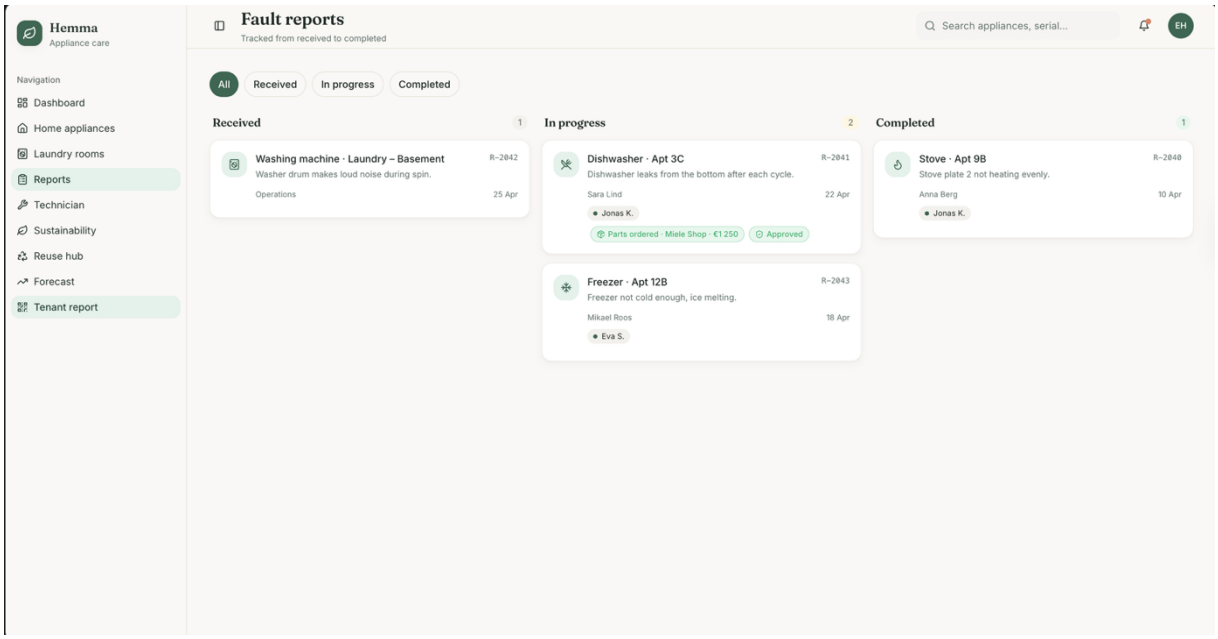


Figure 12 Reports interface showing approved spare-part order

Each report contains detailed information regarding the affected appliance, including appliance type, location, serial number, date of report, and the technician responsible for the task. This structured information supports efficient coordination and enables Operational Managers to make more informed decisions regarding maintenance activities and resource allocation.

Overall, the reporting interface is designed to improve transparency, streamline communication between actors, and reduce the risk of delays or overlooked tasks within the maintenance process. In this way, the functionality addresses several of the challenges related to fragmented information and limited process visibility identified in the empirical findings.

5.1.5 Technician

The system also includes a dedicated interface for technicians, designed to support the execution, coordination and documentation of maintenance tasks. Within this interface, technicians are presented with an overview of assigned reports, including key information such as location, appliance type, serial number, reported issue, and submission date. The interface also supports tasks connected to the reuse hub, where technicians can receive information regarding appliances that need to be collected, transported, installed, or refurbished as part of the reuse workflow (see Figures 13 & 14).

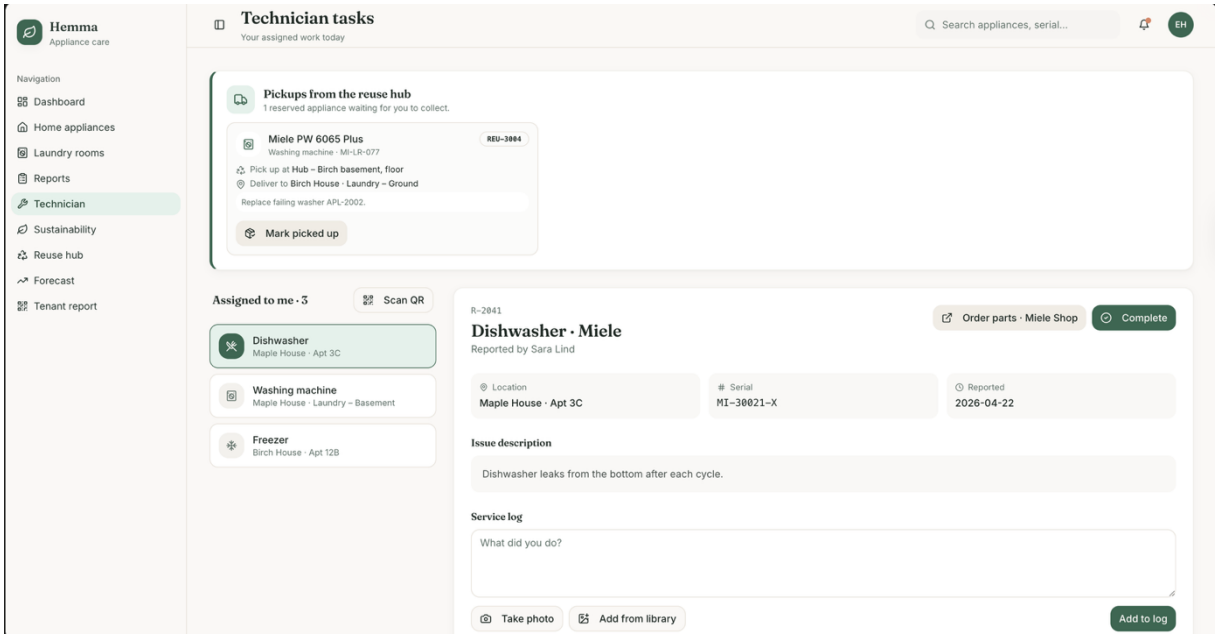


Figure 13 Technician view with expected pickup

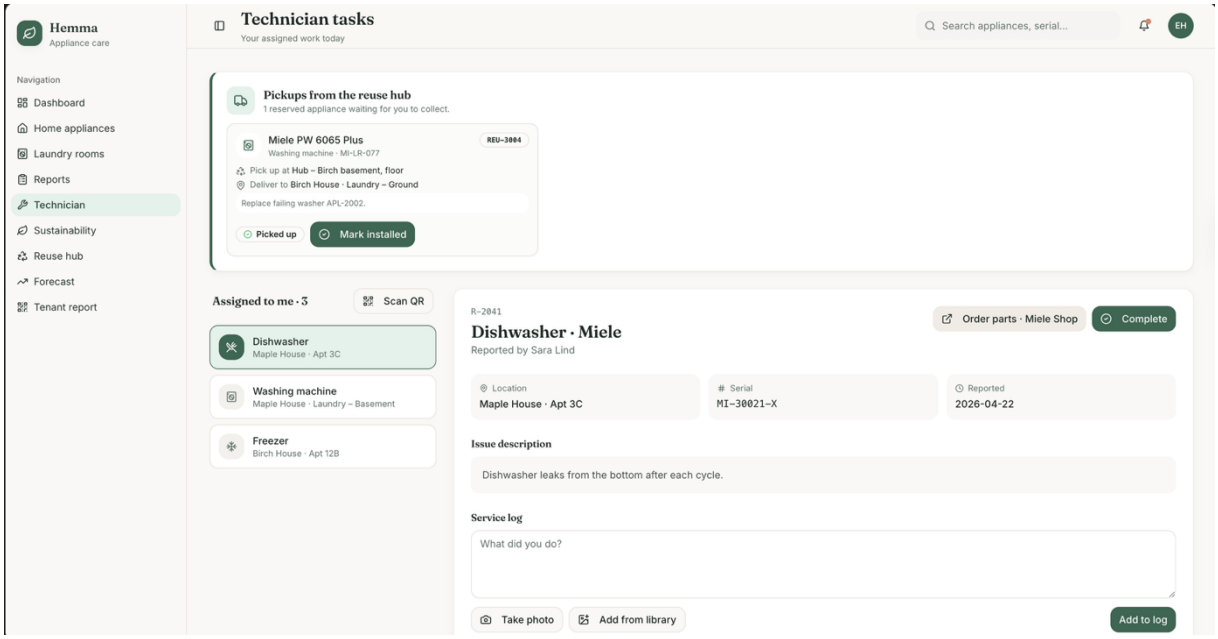


Figure 14 Technician view, appliance marked as picked up

This functionality is intended to improve coordination and simplify the handling of reusable assets throughout the organization. To ensure accurate identification of the appliances, technicians can either select the appliance directly within the system or scan the associated QR code on-site. This reduces the risk of errors and ensures that maintenance activities are linked to the correct asset. During the maintenance process, technicians can also document performed actions by adding notes and uploading images directly into the system. This contributes to improved traceability and creates a structured service history for each appliance.

A central functionality within the technician interface is the integrated spare-part ordering process. Instead of manually updating multiple process stages or status fields, technicians are able to directly access spare-part ordering through links connected to the corresponding manufacturer or supplier of the appliance (see Figure 15).

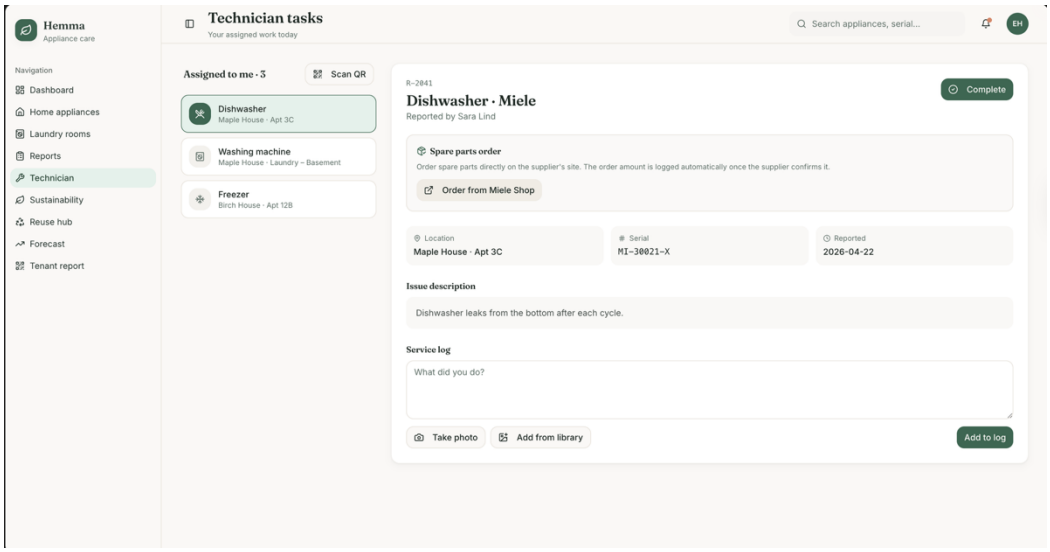


Figure 15 Technician interface showing assigned maintenance tasks

This functionality is intended to simplify the workflow, as technicians are already physically present at the location and can immediately identify, measure, and order the correct components needed for the repair. By allowing technicians to handle the ordering process directly, the system reduces unnecessary administrative work that would otherwise require managers or other employees to manually review and process individual part requests. Once spare parts are ordered, the information is automatically registered within the system and synchronized with the reporting interface used by Operational Managers. Users are therefore able to see which parts have been ordered, the associated costs, and the current progress of the repair process in real time. If the order exceeds a predefined financial approval threshold, the

system automatically prompts the technician to confirm whether managerial approval has been obtained before the order can proceed (see Figure 16 & 17).

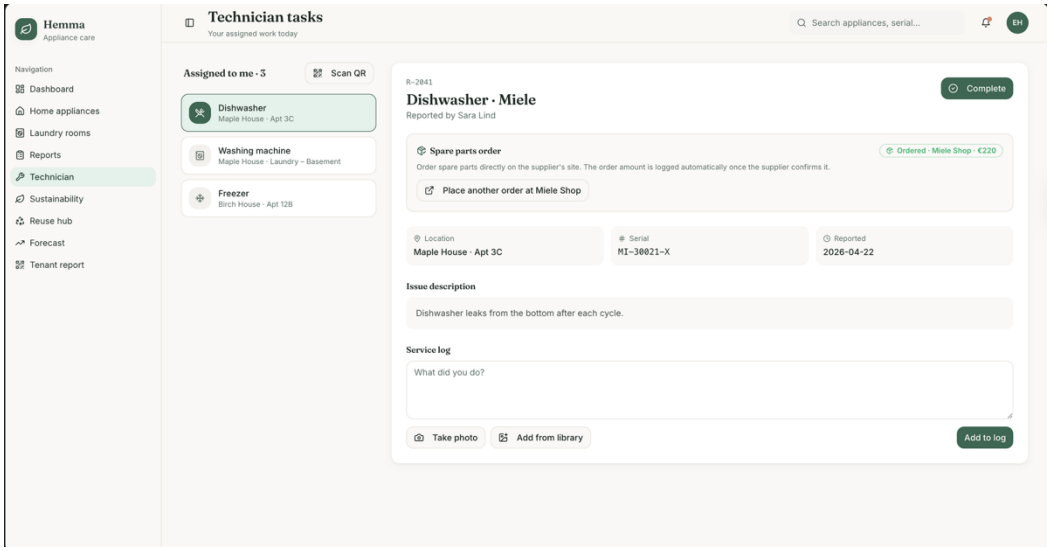


Figure 16 Technician interface showing spare-part order status

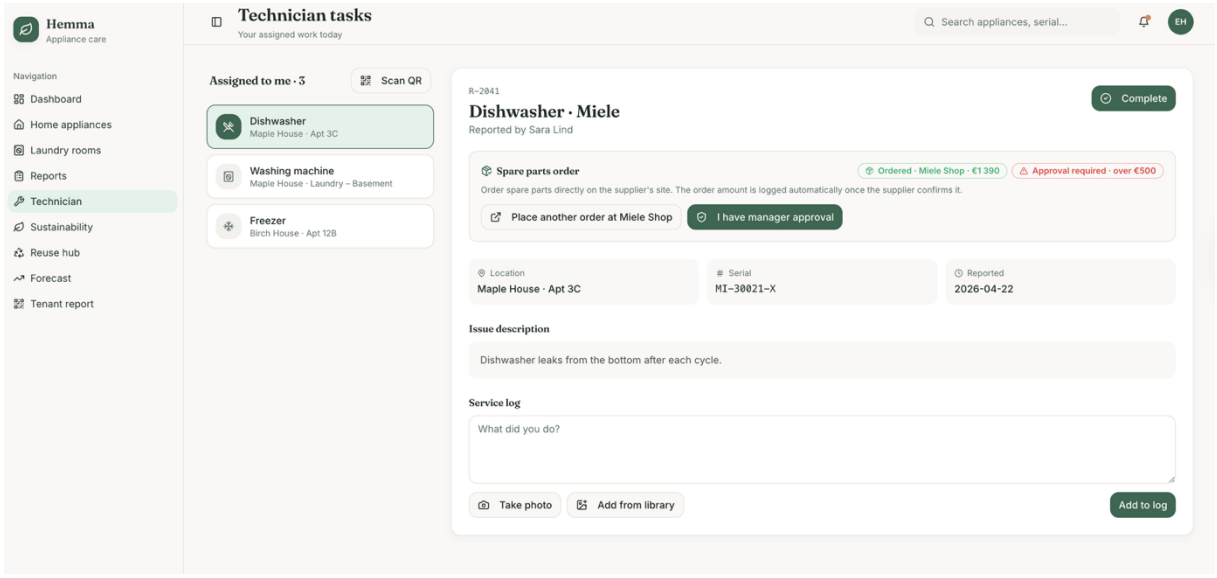


Figure 17 Technician interface showing cost over threshold

This creates a balance between technician autonomy and managerial financial control.

When the maintenance task is completed, the technician marks the report as finished, which automatically updates the system and communicates the status to relevant actors. Overall, the technician interface is designed to support efficient task execution, improve documentation quality, reduce administrative workload, and enhance real-time information flow within the maintenance process.

5.1.6 Sustainability

A key component of the prototype is the sustainability dashboard, which visualizes both the environmental and financial impact of asset-related decisions. The dashboard includes indicators related to reuse, repair, and replacement, together with graphical representations of CO_2 savings and estimated cost reductions over time (see Figure 18). By hovering over visualizations, the user is also able to access more detailed information corresponding to specific time periods and data points (see Figure 19).

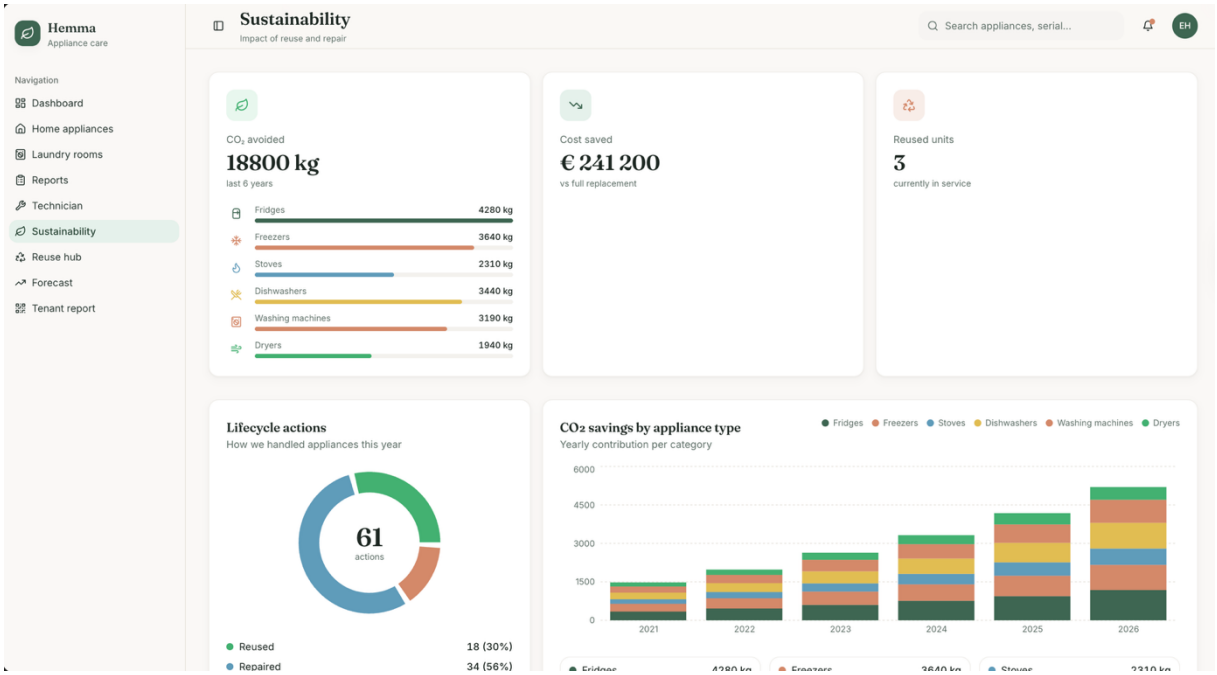


Figure 18 Sustainability dashboard showing environmental and financial indicators

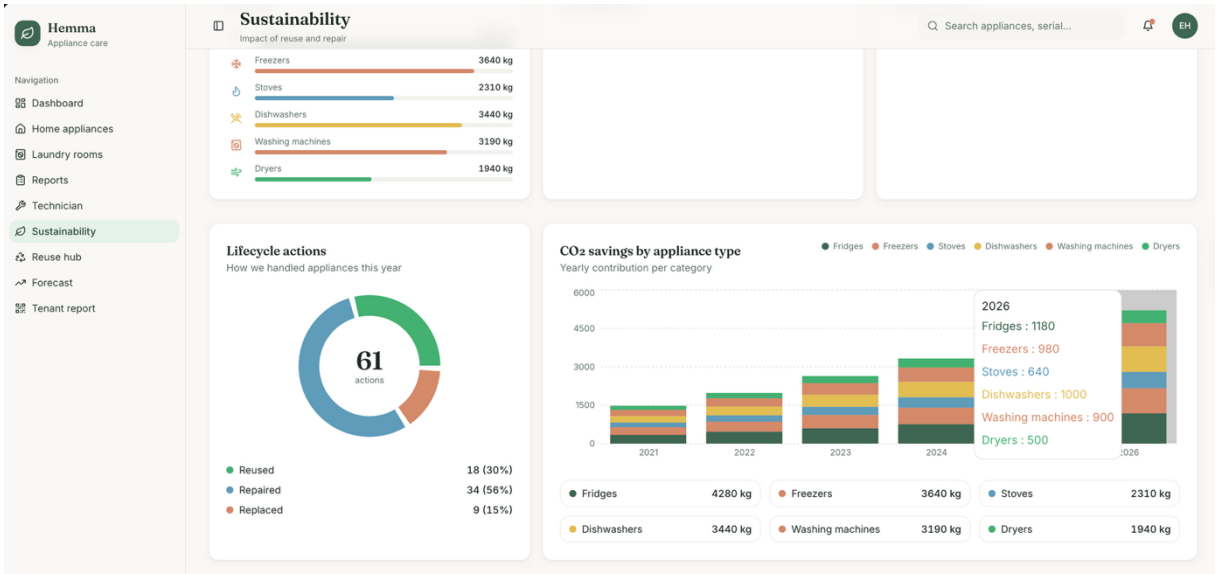


Figure 19 Detailed sustainability data displayed though diagram hover interaction

The purpose of this functionality is to support organizations in understanding, monitoring, and communicating the impact of their asset management practices. By presenting sustainability-related data in a clear and accessible way, the system enables easier reporting and follow-up of environmental performance over time. In addition, the visualization of cost savings highlights the financial benefits associated with extending appliance lifecycles through repair and reuse. This functionality is also intended to support more informed decision-making by making the consequences of different maintenance and replacement strategies more visible. At the same time, the sustainability indicators may function as a motivational element by illustrating the positive environmental and financial outcomes associated with more resource-efficient practices.

Overall, the sustainability dashboard is designed to increase transparency, support sustainability reporting, and strengthen awareness regarding the long-term value of repair, reuse, and lifecycle-oriented asset management. In this way, the prototype illustrates how improved visibility and traceability of asset-related information can contribute to more sustainable and data-driven operational practices.

5.1.7 Reuse Hub

The prototype also includes a dedicated reuse hub, designed to manage appliances that are not currently in use but remain viable for future deployment (see Figure 20). This functionality provides an overview of all appliances stored within the organization's reuse system, thereby supporting more structured and efficient reuse practices. Within the reuse hub, Operational Managers are able to view the total number of available appliances together with their current status. Appliances are categorized based on their readiness for reuse, such as ready for use, requires inspection, or needs refurbishment. This categorization enables users to quickly assess which units are suitable for redeployment and which require additional actions before being reused.

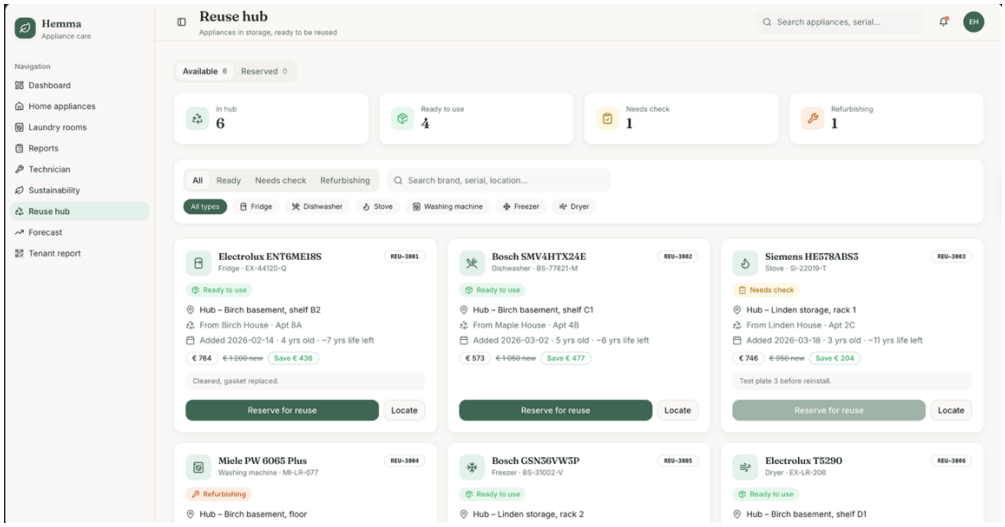


Figure 20 Reuse hub overview showing available appliances and reuse status

The system allows appliances to be filtered and searched based on attributes such as appliance type and status, improving both accessibility and navigation. For each appliance, detailed information is available, including origin, storage location, registration date, estimated age, remaining lifecycle, serial number, and appliance type. In addition, each appliance is assigned an estimated financial and lifecycle-related value based on its condition and remaining useful life (see Figure 20). This value is visualized directly within the reuse hub and functions both as decision support and as an indicator of the financial benefits associated with reuse practices. By clearly displaying the estimated value of reusable appliances, the system aims to reinforce the economic advantages of lifecycle extension and circular resource management.

To facilitate reuse, managers can reserve appliances directly within the system. This action initiates a workflow where the intended destination is specified, and the task is assigned to a technician responsible for relocation or installation (see Figure 21).

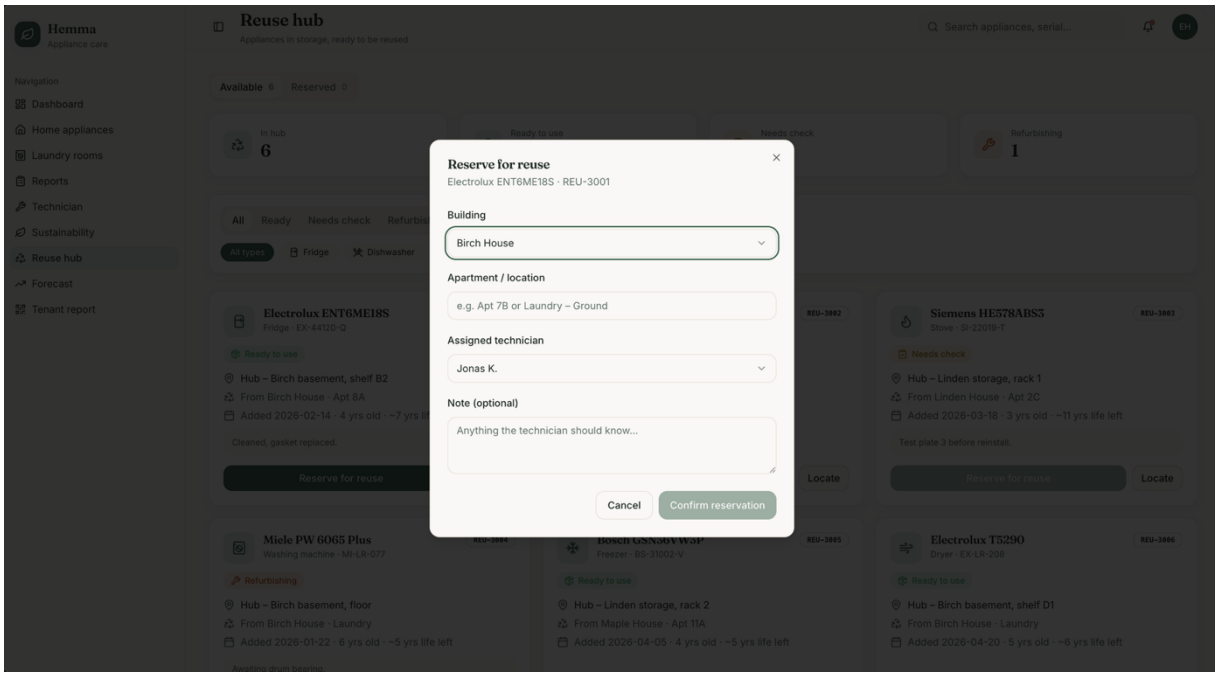


Figure 21 Reuse hub reservation workflow for assigning appliance relocation

Once assigned, the technician receives information regarding where the appliance should be collected, where it should be installed, and its current status of the reuse process (see Figure 13). Managers are also able to monitor whether the appliance has been collected or fully installed. Reservations can additionally be modified or cancelled when necessary, ensuring flexibility within the workflow. Once a reused appliance has been successfully installed, the estimated lifecycle and financial value associated with the appliance are automatically transferred into the forecasting and sustainability functionalities of the system. This updates both projected cost savings and sustainability metrics connected to reuse activities, thereby improving visibility of the long-term financial and environmental impact of circular asset management practices.

The reuse hub is based on the principle that appliances are registered into the system at the point of storage, for example through QR code scanning. This ensures that appliances entering

the hub are immediately integrated into the system and available for tracking and future deployment.

Overall, the reuse hub is designed to support circular asset management by improving visibility of available resources, enabling structured reuse, supporting operational coordination, and reducing unnecessary procurement of new appliances.

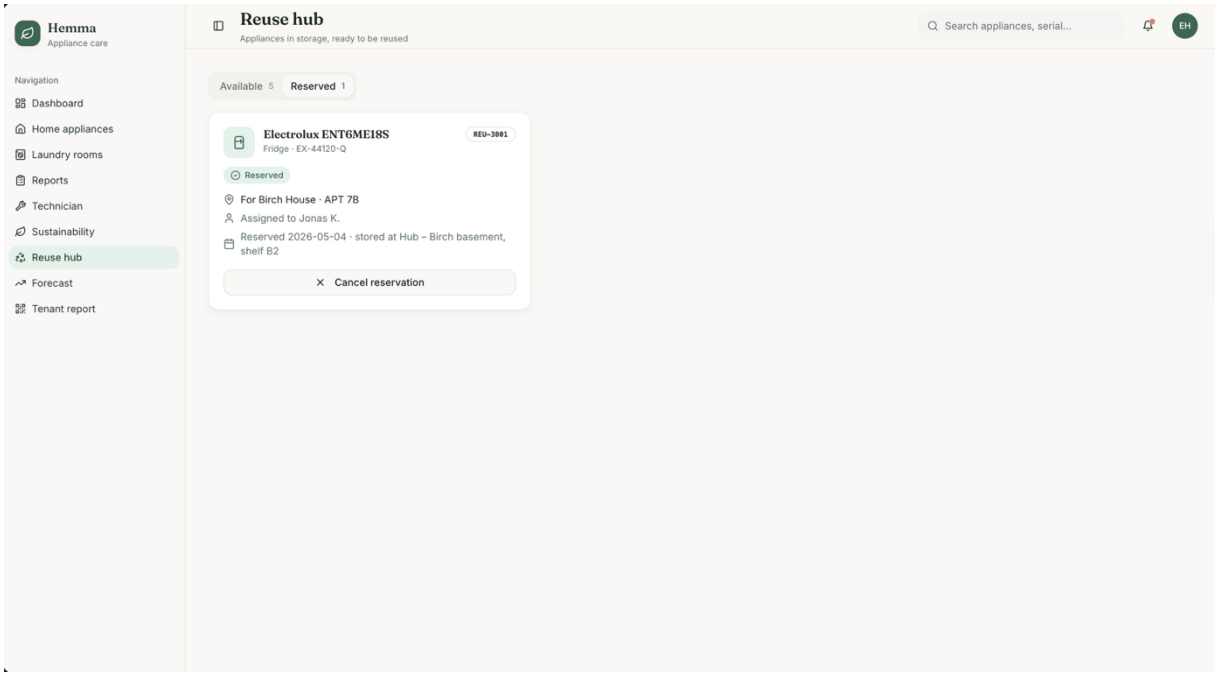


Figure 22 Reserved appliances overview in reuse hub

5.1.8 Forecast

The prototype also includes a dedicated forecasting page designed to support long-term planning and financial estimation related to appliance management. This functionality enables users to project future expenses over different time horizons, including 3, 5, and 10 years.

The forecasts are divided into key operational categories, such as service and maintenance costs, as well as appliance replacement expenses. In addition, the system provides an annual overview, where projected expenditures are visualized across different appliance categories, including refrigerators, freezers, stoves, dishwashers, washing machines, and dryers (see Figures 23 & 24). This allows users to identify cost distribution patterns and gain a clearer understanding of which asset categories contribute most to overall expenditure.

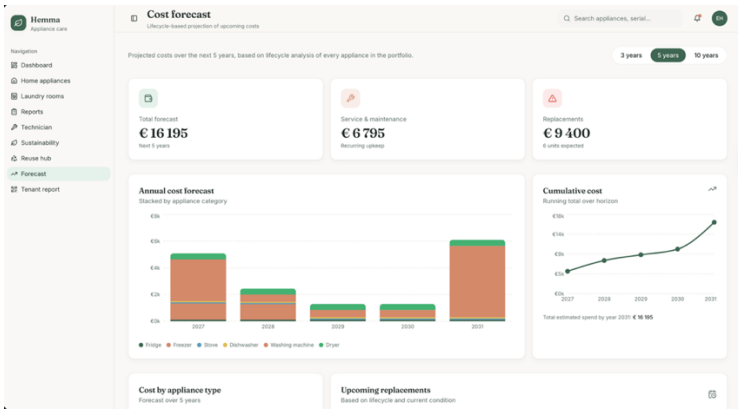


Figure 23 Forecast overview showing projected appliance-related costs

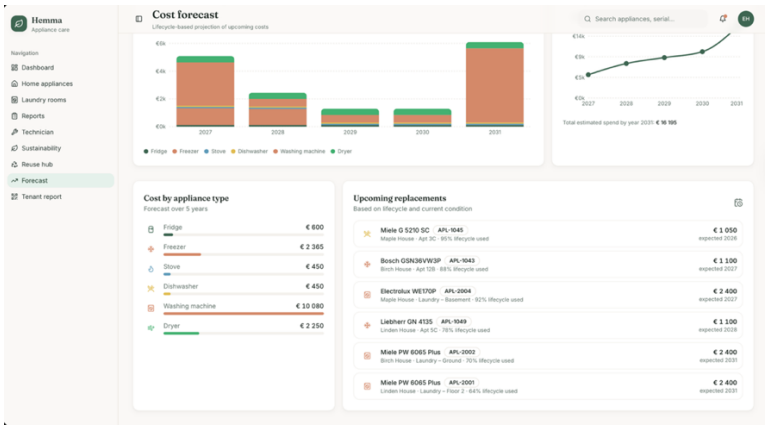


Figure 24 Lower section of forecast overview

To further support planning and resource allocation, the system includes a cumulative cost visualization showing estimated spending over time. This functionality enables users to identify trends, anticipate future financial needs, and support more proactive budgeting processes. A complementary view additionally presents forecasts grouped by appliance type, providing a more detailed breakdown of expected expenses (see Figure 24).

The forecasting page also highlights upcoming replacement needs by identifying which appliances are expected to require replacement, when replacements are likely to occur, and the estimated associated costs. This provides a clearer basis for procurement planning, resource allocation, and the estimated associated costs. Overall, the forecasting functionality is designed to improve financial visibility, support proactive and data-driven decision-making, and reduce uncertainty within long-term asset management planning.

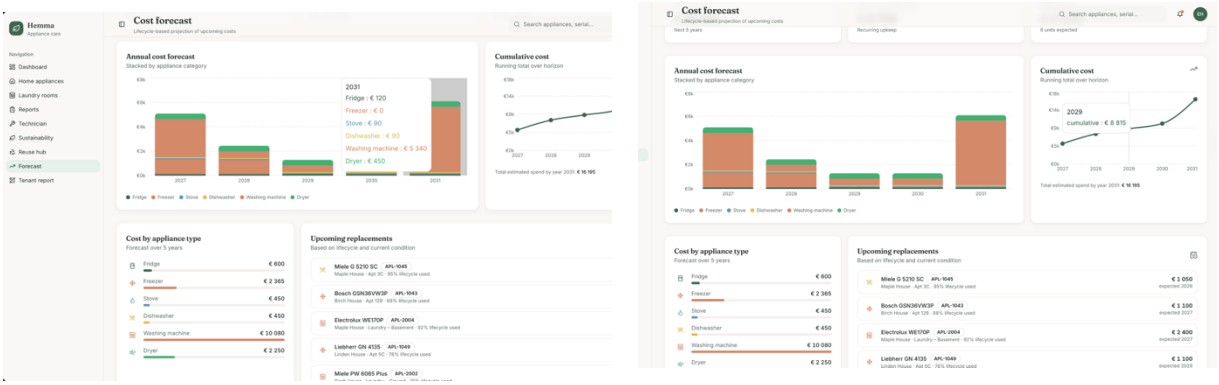


Figure 25 Detailed forecasting data displayed though diagram hover interaction

5.2 Prototype Evaluation

To evaluate the proposed solution, follow-up interviews were conducted with selected respondents, including the Operational Manager (Large Real Estate), Stefan Hvalgren (Fixkod), and a tenant within a Swedish real estate context. The purpose of these interviews was to gather feedback regarding the prototype's usability, functionality, perceived value, and practical relevance within property management operations. During the interviews, the prototype was presented to the participants, and respondents were guided through its different functionalities and workflows. Participants were encouraged to reflect on how the system could potentially be integrated into their daily work practices, what operational value it could contribute, and what limitations or implementation challenges they perceived.

The interviews followed an exploratory and open-ended structure, allowing respondents to freely discuss their experiences, opinions, and suggestions for further development. Focus was placed on how the prototype addressed previously identified challenges related to fragmented information, limited traceability, lifecycle visibility, maintenance coordination, and decision-making support. Overall, the feedback indicated a positive perception of the prototype and highlighted its potential value in improving information accessibility, traceability, lifecycle management, and decision-making processes within property management operations. Respondents described the system as structured, easy to understand, and potentially applicable for both smaller and larger real estate organizations. Several respondents particularly emphasized the value of improved overview regarding warranties, lifecycle status, maintenance history, and reuse coordination.

The collected feedback was analyzed using a focused thematic approach, where responses were categorized into recurring themes related to perceived value, usability, implementation challenges, organizational fit, and suggested improvements. The insights gathered from the evaluation process were subsequently used to refine and further develop the final version of the prototype presented in this study.

5.2.1 Dashboard Feedback

One suggestion that emerged during the feedback interviews concerned the visualization of sustainability impact within the sustainability dashboard (see Figure 26). One of the respondents proposed that sustainability indicators could be presented through more intuitive and engaging visual elements, such as trees saved, marine life protected, or similar symbolic indicators, rather than relying solely on traditional graphs and numerical values. According to the respondent, this type of visualization could make the environmental impact easier to understand at a glance while also creating a stronger emotional connection to the sustainability efforts.

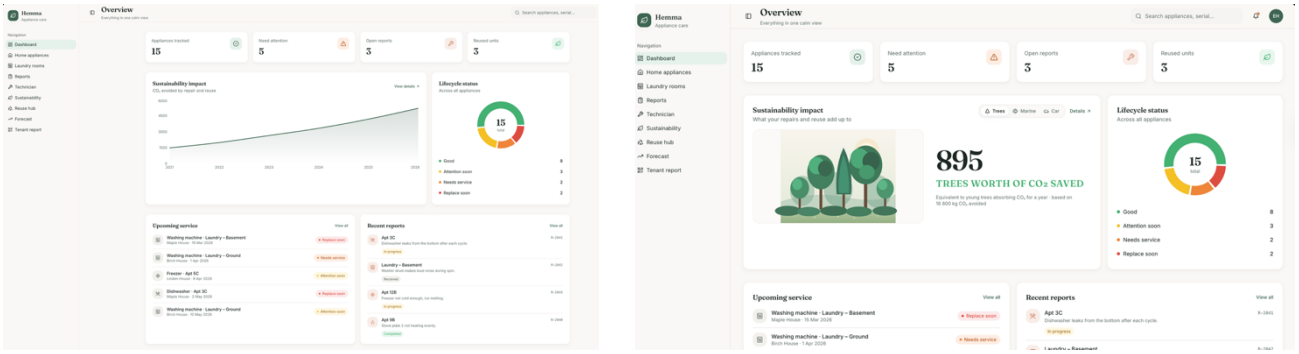


Figure 26 Dashboard, before (Left) and after changes (Right)

As a result of this feedback, the prototype was further developed to include more visually engaging sustainability indicators on the dashboard start page. The purpose of these additions was to create a more motivating and accessible overview of the organization's environmental impact while still complementing the more detailed sustainability statistics and forecasting functionalities available in the sustainability section of the system.

5.2.2 Improvements Related to Laundry Room and Technician Workflow

Additional suggestions for improvement also emerged during the follow-up interviews. Within the laundry room functionality, the respondents proposed integrating runtime tracking directly into technician workflows. By requiring technicians to register runtime during service visits, the system could support more accurate maintenance planning and lifecycle-based service intervals (see Figure 27).

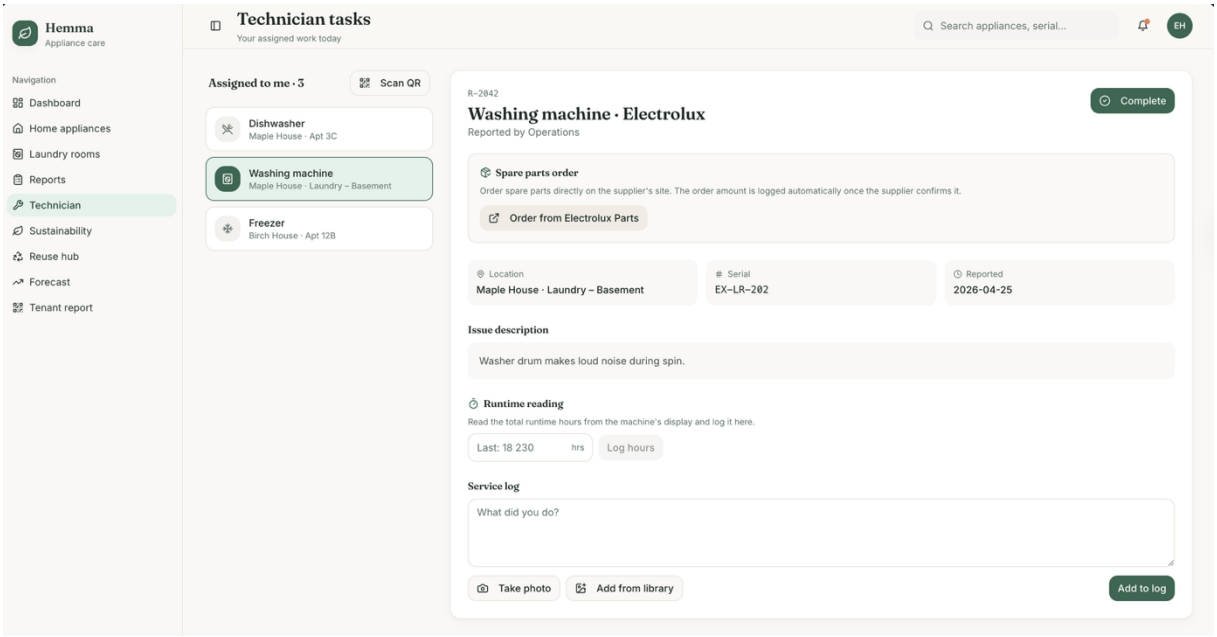


Figure 27 Technicians view with updated runtime reading

The technician workflow was also perceived positively, particularly the possibility of integrating spare-part ordering directly into the system. This functionality was viewed as beneficial for reducing administrative workload while still maintaining managerial financial control through approval thresholds for larger purchases. Respondents emphasized that instead of manually updating work statuses through drop-down menus, as implemented in the earlier version of the prototype, status updates could instead occur automatically through the spare-part ordering process itself (see Figure 27). For example, once a technician places an order for spare parts, the system would automatically notify the manager while simultaneously updating work status accordingly. This was perceived as a more intuitive and simplified workflow, as the ordering process itself implicitly indicates that the repair is awaiting parts.

Respondents further highlighted that allowing technicians to directly order spare parts would significantly simplify daily operations by reducing the need for managers to manually review and approve every minor request. At the same time, the respondents emphasized that importance of maintaining predefined financial approval thresholds, meaning that technicians would still require managerial confirmation for purchases exceeding a predetermined cost limit. One example discussed during the interviews was that repair costs for reused appliances should not exceed approximately half of the price of purchasing a newly manufactured appliance.

Another suggestion concerned the Estimated Time of Arrival (ETA) dropdown functionality that existed in the first version of the prototype. This feature was perceived as somewhat unnecessary and overly complex, since the spare-part ordering itself already communicates delays or waiting times within the workflow. Consequently, respondents suggested that the ETA functionality could be replaced entirely by the integrated ordering system (see Figure 28). The respondents also recommended that the spare-part ordering interface should provide direct access to the corresponding manufacturer or supplier connected to each appliance. This

was viewed as an important way to further simplify procurement processes and improve collaboration between property organizations and external suppliers.

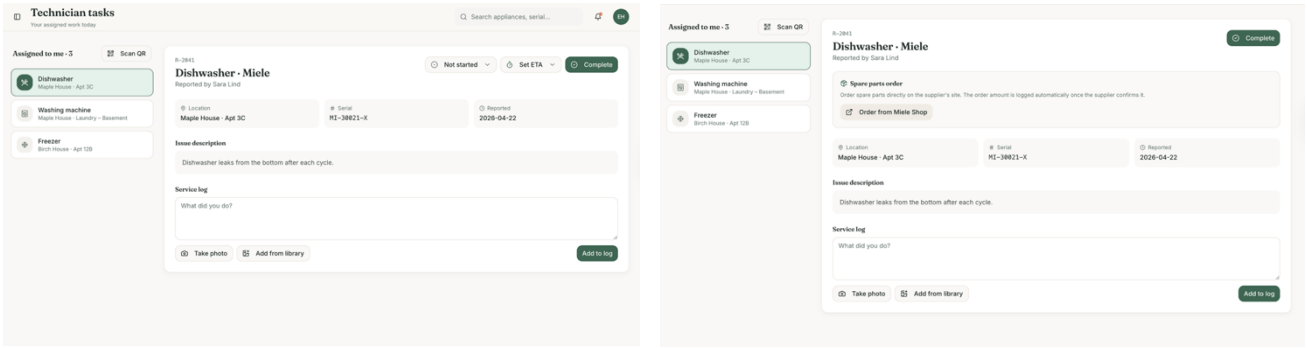


Figure 28 Technician view, before changes (Left) and after changes (Right)

5.2.3 Improvements Related to Sustainability Data

The sustainability functionality also received generally positive feedback, although respondents proposed improvements related to the structure and presentation of sustainability data. One important recommendation was to separate environmental indicators across different appliance categories in order to provide clearer visibility of how specific product families contribute to CO_2 emissions and environmental savings (see Figure 29).

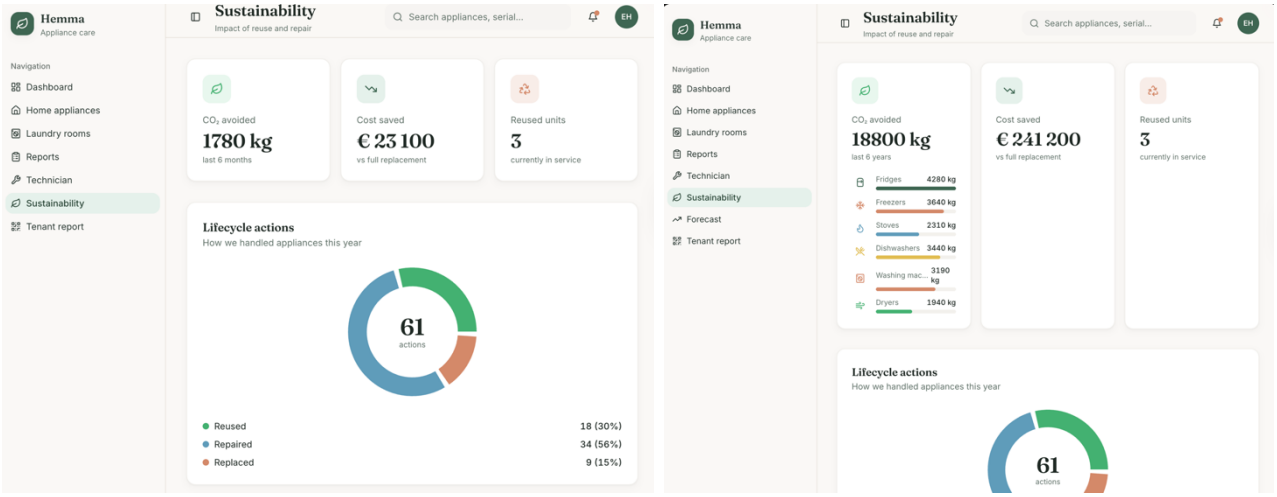


Figure 29 Sustainability view before the Changes to the CO2 indicator (left), and after changes (right)

Respondents also emphasized the importance of clarifying that the estimated environmental savings primarily represent avoided emissions with the manufacturing of new appliances

when existing appliance lifecycles are extended through repair and reuse practices. This clarification was considered important to improve the understanding of how the sustainability calculations should be interpreted. In addition, the original monthly sustainability reporting was perceived as less useful than yearly visualization, since sustainability metrics are generally evaluated over longer periods of time. As a result, the sustainability dashboard was adjusted to emphasize yearly trends and long-term environmental impact more clearly.

5.2.4 Financial Forecasting Functionality

The financial forecasting functionality was perceived as particularly valuable from a managerial perspective. Respondents highlighted that visualizations of projected costs across different appliance categories could support budgeting, long-term planning, and decision-making regarding future replacement needs. Similarly, the sustainability dashboard and cost-saving indicators were viewed as important decision-support tools, since they visualized both environmental and financial benefits associated with reuse practices and lifecycle extensions. Respondents explained that this could make it easier to justify investments in reuse systems, preventive maintenance practices, and additional organizational resources required to support circular asset management processes.

5.2.5 Lifecycle and Warranty Indicators

Feedback from the follow-up interviews also resulted in the addition of new lifecycle-related indicators within the system. In particular, the warranty indicator was directly integrated into the lifecycle visualization, allowing users to clearly see whether an appliance remained covered under warranty together with the estimated remaining warranty period. This functionality was perceived as valuable decision support when evaluating whether appliances should be repaired, reused, or replaced. Respondents explained that easier access to warranty-related information could contribute to faster and more informed maintenance decisions while simultaneously reducing uncertainty within repair and replacement processes.

5.2.6 Reuse Hub Functionality

The reuse hub functionality was also perceived as highly relevant by the respondents. The Operational Manager (Large Real Estate) particularly appreciated the possibility of tracking which appliances were available, reserved, installed, or undergoing refurbishment within the reuse workflow. According to the respondent, having a centralized overview of reusable appliances could improve both operational coordination and long-term lifecycle management. Respondents also emphasized the importance of assigning financial and lifecycle-related values to reused appliances when they are registered into the system. This was considered as

particularly important to avoid overestimating the remaining value or expected lifespan of older appliances. As a result of this feedback, additional functionality was introduced into the prototype to more clearly visualize the registered financial value and estimated remaining lifecycle of appliances within the reuse hub (see Figure 30 & 31). This addition also contributed to making the financial benefits of reuse more visible, thereby reinforcing the intended purpose of the system by illustrating how lifecycle extension and reuse can contribute to cost savings.

In addition, the possibility of directly connecting reuse activities to forecasting and sustainability reporting was perceived as highly beneficial for both operational planning and long-term asset management. By linking reused appliances to financial forecasting and sustainability metrics, the system was considered capable of providing clearer decision support regarding future replacements, resource allocation, and environmental impact.

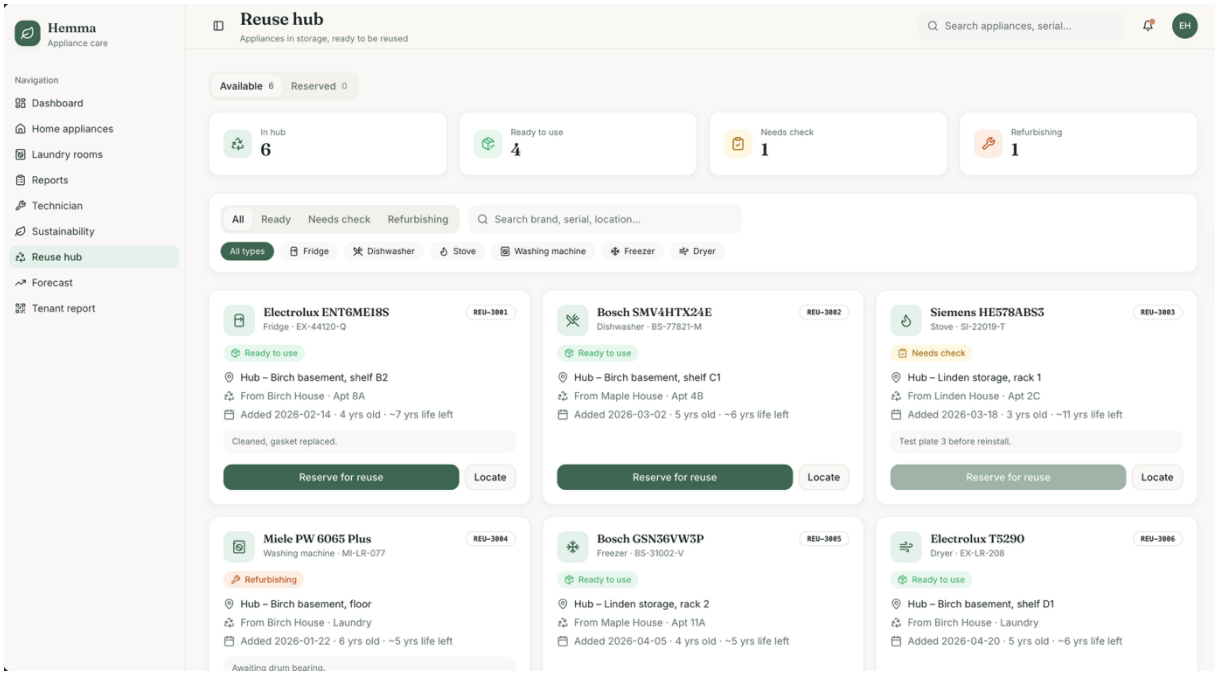


Figure 30 Reuse hub, before changes

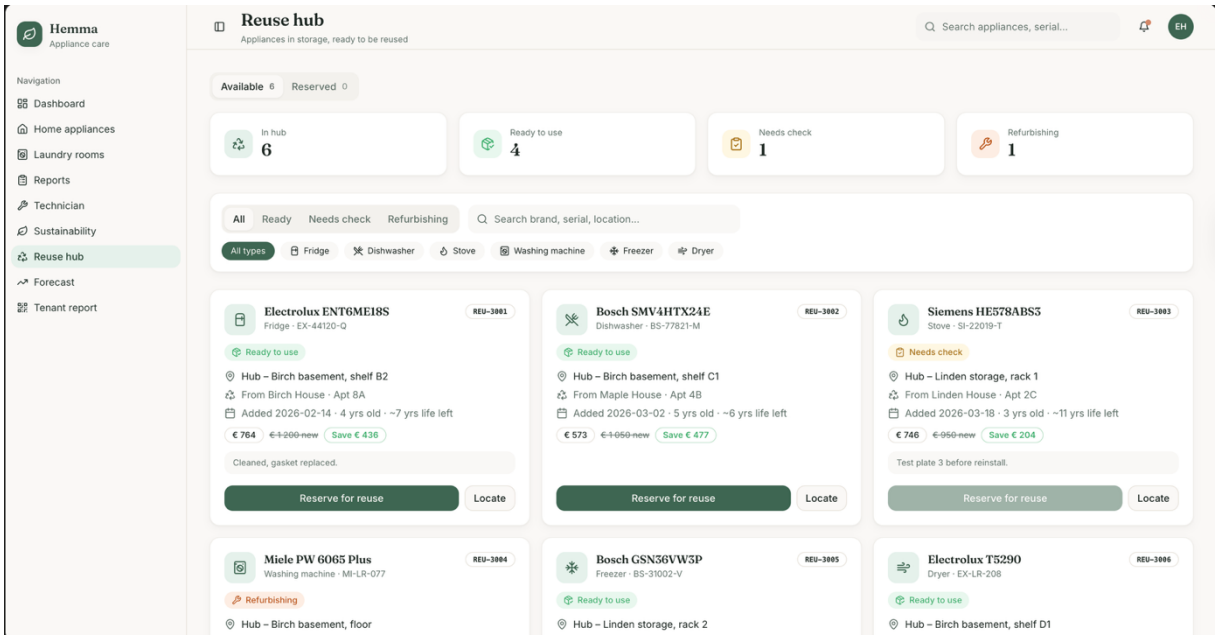


Figure 31 Reuse hub, after changes with clear values

5.2.7 Tenant Perspective

The tenant perspective was evaluated separately, focusing specifically on the reporting functionality that tenants would realistically interact with during daily use. Overall, the tenant response to the prototype was highly positive, particularly regarding the simplicity and accessibility of the fault reporting process. The tenant emphasized that fault reporting often occurs in situations where users are already frustrated due to malfunctioning appliances, making it especially important that the reporting process remains simple, intuitive and easy to understand.

"*Since it's usually something negative, it is nice not to have to handle a long process. It is the worst to tackle a complicated and hard-to-understand system when you are already in a negative mindset." – Tenant in a Swedish real estate context.*

The QR-based reporting functionality was also appreciated, as it enabled tenants to quickly identify and report issues connected to a specific appliance. At the same time, the tenant emphasized the importance of offering alternative interaction methods, such as selecting appliances manually through the interface, since not all users may have access to QR-scanning functionality or feel comfortable using it. The confirmation step after submission was also perceived positively, as it reduced uncertainty regarding whether the issue had been successfully registered and processed by the organization.

The remaining respondents were also presented with the tenant interface, and the feedback was similarly positive. Respondents agreed that the simplicity of the reporting process aligned well with the needs and expectations expressed by the tenant participant. In particular, they appreciated how quickly and easily a tenant could report an issue without needing to navigate

through complex systems or workflows. Respondents also viewed it positively that the reporting functionality could either be directly connected to the prototype itself or integrated with existing organizational fault reporting systems or email-based workflows. This flexibility was considered important, as different organizations may already use established reporting systems that they prefer to maintain while still benefiting from the simplified QR-based access and traceability functionality provided by the prototype.

Overall, the feedback interviews suggest that the prototype addresses several of the operational and informational challenges identified throughout the empirical findings. At the same time, respondents emphasized interviews highlighted that successful implementation would depend on factors such as integration with existing systems, clear registration and lifecycle-management processes, structured change management, and organizational willingness to maintain consistent information management practices.

6. Analysis and Discussion

This chapter presents the analysis and discussion of the empirical findings in relation to the theoretical framework and the developed prototype. The chapter analyzes how challenges related to information visibility, traceability, lifecycle management, and organizational factors influence decision-making and sustainable resource utilization within property management contexts.

6.1 Lack of Integrated Asset Information

The empirical findings indicate that limited data visibility and information accessibility are central challenges to reach an overarching and coherent overview of the organizational appliances and their associated information, such as warranties, lifecycle status, and service history (Operational Manager, Large Real Estate; Stefan Hvalgren, Fixkod). At the same time, the findings suggest that access to data alone is not sufficient; information must also be structured, centralized, and easily accessible in order to support effective decision-making. This aligns with Berisha-Shaqiri (2014), who emphasizes that effective decision-making depends on access to accurate, timely, and relevant information. In the absence of such information, the organization risks fragmented data structures, poor communication, and reduced decision quality.

When managing physical assets such as appliances, information becomes particularly important throughout the asset lifecycle. According to Diop et al. (2021), decisions made already during the early lifecycle stages, such as installation and registration, influence long-term operability, maintenance efficiency, and sustainability outcomes. If reliable lifecycle

information is missing or inconsistently managed, organizations' risk ineffective maintenance strategies. limited traceability, and premature replacement of assets.

The empirical findings illustrate several examples of these challenges in practice, The Operational Manager (Large Real Estate) explained that asset-related data is stored within a centralized system. However, Stefan Hvalgren (Fixkod) highlighted that even when data exists within the organization, it is often fragmented across multiple systems, communication channels or informal workflows, making it difficult to access and utilize in practice. One example discussed during the interviews was that maintenance, warranties or service information may be communicated through emails or informal communication channels rather than being consistently registered within the central system. In addition, if an appliance is replaced and its identifier is lost or the newly installed appliance is not properly registered, it becomes difficult to determine which future appliance reports or maintenance activities refer to. As a result, important historical information risks being lost over time, limiting both overview and future planning capabilities. In larger organizations, these information gaps may initially appear manageable, as individual missing data points are not always perceived as critical. However, over time, the accumulation of fragmented or missing information may negatively impact both operational decision-making and the organization’s ability to manage assets efficiently and sustainably.

Taken together, both empirical findings and the previous research suggest that the core issue is not necessarily the absence of information, but rather the lack of structured, integrated, and easily accessible asset-related data. This contributes to inefficiencies in daily operations and complicates decisions regarding whether appliance should be repaired, reused, or replaced. From this perspective, digital traceability solutions, such as the QR-based prototype, can be understood as a potential way of addressing these challenges. By centralizing information and directly connecting it to individual assets, such systems can improve visibility, reduce information loss, and support more effective lifecycle-oriented management of asset-related information over time.

6.2 Consequences of Limited Visibility for Decision-Making

While the previous section highlighted issues related to fragmented and inaccessible data, this section focuses on how these limitations affect organizational decision-making processes.

The literature clearly suggests that limited access to relevant information often contributes to poorer decision-making outcomes. One central aspect is the increased uncertainty associated with decision-making, as emphasized by Schultz et al. (2010). Although uncertainty is an unavoidable component of decision-making, particularly in situations involving risk, uncertainty is significantly amplified when accessible and reliable information is lacking. This becomes especially problematic when critical information related to assets, maintenance history, or lifecycle status is missing. In situations involving operational risk, access to relevant information regarding the variables influencing the outcome becomes essential. At

the same time, Schultz et al. (2010) emphasize that collecting additional information may be costly, time-consuming, and practically difficult. In this context, centralized systems where information is structured and easily accessible may play an important role in reducing the complexity in both small operational decisions and larger strategic planning processes, while simultaneously contributing to long-term efficiency.

As highlighted by Diop et al (2021), relevant information should be established early in the asset lifecycle. Even information that may initially appear insignificant, such as a serial number or installation details, may become extremely difficult and costly to retrieve later (Tsay et al., 2022). This becomes particularly relevant in within property management contexts, where assets are often physically distributed inside tenants' homes, making direct access difficult (Stefan Hvalgren, Fixkod). Under such situations, decisions are more likely to be based on experience, assumptions or simplified assessments rather than structured and reliable data. A clear example in the empirical findings is that functional appliances are sometimes replaced despite still having remaining lifecycle value, particularly during larger renovation projects (Stefan Hvalgren, Fixkod). According to respondents, this is often justified by the difficulty of accessing reliable information, making replacement the easier and less risky alternative.

Furthermore, the analysis indicates that limited visibility contributes to risk-minimizing organizational behavior. Organizations may avoid reusing appliances when information regarding condition, maintenance history, or expected remaining lifespan is unclear, since uncertainty regarding future functionality may lead to additional costs, increased maintenance needs and dissatisfied tenants. If a reused appliance fails shortly after installation, the organization risks both financial losses and reduced customer satisfaction. Conversely, limited information may also result in organizations investing resources into repairing appliances that have already reached the end of their practical lifecycle. In such cases, time, labor, and financial resources may be allocated toward actions that do not contribute to long-term value creation. These findings align with Berisha-Shaqiri (2014), who emphasizes that effective decision-making depends on access to accurate, relevant and timely information. Without this, information systems risks becoming fragmented, communication deteriorates, and decision quality decreases. This becomes particularly problematic in asset-intensive environments, where data is often extensive, distributed, and difficult to manage effectively. The empirical findings further illustrate that even organizations already utilizing centralized information systems still experience information gaps. The Operations Manager (Large Real Estate) explained that asset-related information is continuously updated within the existing system. However, gaps still emerge, particularly in connection with replacement processes, where information regarding previous assets may be lost or inconsistently registered. This contributes to uncertainty and inefficiencies within operational processes.

In addition, limited data visibility affects an organization's ability to work proactively, particularly in relation to forecasting and long-term planning. Without a clear overview of assets and their lifecycle status, it becomes difficult to anticipate future replacement needs or plan procurement activities efficiently over time (Stefan Hvalgren, Fixkod). As a result, organizations' risks increased uncertainty, overprocurement, and inefficiencies within

operational planning processes. In summary, the analysis indicates that access to structured, centralized, and easily accessible information is essential for enabling informed decision-making, improving resource utilization, and supporting long-term value creation in line with a life cycle perspective (Diop et al, 2021). From this perspective, the findings highlight the importance of systems capable of reducing uncertainty by providing accessible, structured, and continuously updated asset-related information in real time.

6.3 Reactive and Risk-Minimizing Decision Behavior

As discussed in the previous section, uncertainty and limited access to information have a significant influence on how decisions are made within asset management processes. This section further explores how uncertainty contributes to more reactive and risk-minimizing organizational decision-making behavior.

The empirical findings show that decisions regarding appliances are often made on a case-by-case basis. According to the Operational Manager (Large Real Estate*),* decisions concerning whether appliances should be repaired or replaced are primarily based on factors such as cost, perceived condition, and assessments provided by technicians. The respondent further explained that technicians are typically sent out to inspect the appliance before reporting their evaluation back to the management, which subsequently makes the final decision.

However, when information is incomplete or difficult to access, these decisions tend to become simplified or "rounded up" in order to reduce uncertainty. Stefan Hvalgren (Fixkod) explained that many organizations within the sector perceive it as difficult and time-consuming to fully utilize the complete lifecycle of appliances. As a result, replacement is often perceived as the easier and safer alternative compared to repair or reuse. This becomes particularly problematic when organizations lack a clear overview of the existing assets and their condition. Without easily accessible information regarding appliance history, remaining lifecycle or location, it becomes difficult to coordinate reuse across different properties. Consequently, appliances that still retain functional value risk remaining unused or prematurely discarded.

At the same time in the findings demonstrate that organizations already work with internal reuse solutions during larger renovation projects, where appliances are collected and stored in reuse hubs. The Operational Manager (Large Real Estate) described this practice as a way of extending the lifecycle of the functioning products. However, the analysis suggests that simply storing appliances is not sufficient if the organization lacks the information required to effectively evaluate, track and reuse them. As Stefan Hvalgren (Fixkod) pointed out, storage itself also generates costs, particularly when products remain unused for extended periods of time, while simultaneously increasing the risk of overstock and inefficient resource utilization.

This highlights the importance of structured information and accessible information as a foundation for effective decision-making. Stefan Hvalgren (Fixkod) explained that what is

currently missing is a system capable of effectively tracking which appliances have exceeded their lifecycle and which remain suitable for continued use. Although some organizations already possess systems that partly support this work, respondents emphasized that it is still difficult to quickly and easily access the necessary information at the right time. This can also be connected to statements from the Operational Manager (Large Real Estate), who described an ideal scenario where all information connected to an apartment and its appliances would be directly accessible within the system, including installation date, warranty information, maintenance history and location. This suggests that the existing system do not fully support the operational needs of the organization, despite already being used in practice.

These findings align with Tsay et al. (2022), who identify information accessibility as one of the major challenges within facility management. According to their research, facility managers often lack fast and effective access to the information required to manage maintenance tasks and work orders, making it more difficult to make informed and timely decisions. This issue is similarly reflected in the empirical findings, where replacement of appliances may result in identifiers or historical information being lost, thereby creating inefficiencies and uncertainty within operational processes. Furthermore, Tsay et al. (2022) and Olimat et al. (2023) emphasize that information often exists but is stored in fragmented formats, such as separate files, documents, or systems, making information management both inefficient and costly. Manual registration of information additionally increases the risk of human errors and inconsistencies. This can be connected to Stefan Hvalgren (Fixkod), who explained that organizations could theoretically manage this information manually, but that such an approach would become excessively time-consuming and economically unsustainable over time. The analysis can also be connected to Razak et al. (2023), who describe traceability as the ability to track and trace a product's history, location, and status throughout its lifecycle. According to their research, traceability contributes to improved risk awareness, increased reliability, and enhanced trust between stakeholders. From this perspective, the lack of traceability identified in the empirical findings contributes directly to uncertainty and reinforces more reactive decision-making processes.

Another aspect identified in the findings is the perceived cautiousness within the public sector organizations. According to both Stefan Hvalgren (Fixkod) and Hampus Granström (Project Leader Internationalization), organizations within the public sector tend to prioritize safer and more predictable decisions, particularly in situations characterized by uncertainty. This may involve “rounding up” procurement volumes or prematurely replacing appliances in order to minimize the risk of future operational problems. This partially aligns with the common assumption that Nicholson-Crotty et al (2019) who note that public sector organizations are often perceived as more risk-averse than the private sector. At the same time, Nicholson-Crotty et al. (2019) emphasize that such behavior is highly context-dependent and influenced by how consequences and risks are framed. Nevertheless, the findings of this study suggest that uncertainty regarding future failures, dissatisfied tenants, and operational disruptions contributes to more cautious decision-making behavior within property management context. A clear example of this is the difficulty of forecasting future needs. Respondents described situations where organizations choose to purchase larger quantities of appliances than initially

required, since uncertainty regarding future needs may otherwise result in delays, shortages or operational disruptions later in the process.

From this perspective, improved traceability and easier access to structured information could contribute to more proactive and data-driven decision-making. Through increased visibility of asset conditions, lifecycle status and future replacement needs, organizations may reduce uncertainty, improve forecasting capabilities, and simultaneously support more effective and sustainable resource utilization.

6.4 Barriers to Effective Lifecycle and Circular Management

As discussed in previous sections, limited visibility and restricted access to information create several challenges for organizations attempting to establish effective lifecycle-oriented and circular asset management practices. This section, therefore, focuses on the barriers that affect the possibility of creating more sustainable and long-term lifecycle and reuse processes.

Without relevant and accessible information, it becomes significantly more difficult to determine whether an appliance remains suitable for continued use and how much of its expected lifecycle remains. The Operational Manager (Large Real Estate), for example, explained that if a refrigerator compressor stops functioning and the appliance is already around eight to ten years old, it is often considered more feasible to replace the entire appliance rather than repair it. However, for such decisions to be made effectively, information regarding the appliance's age, maintenance history, condition, and expected remaining lifecycle must be available and easily accessible. If this information is missing, establishing an effective reuse system becomes considerably more difficult. Respondents described the situations where fully functional appliances are not reused despite still having remaining functional value. The Operational Manager (Large Real Estate) also emphasized that technicians require access to relevant information regarding appliance history and lifecycle status in order to support consistent decision-making processes. Otherwise, decisions risks becoming inconsistent, for example if a technician recommend replacement while management determines that the appliance should remain in use. At the same time, this division of responsibilities can also be viewed as a potential strength when accurate information exists within the system. In such cases, technician assessments can function as a complement to historical and lifecycle-related information, thereby supporting more informed decisions regarding repair, reuse or replacement activities.

The empirical findings further show that the organization utilizes a reuse hub where older appliances are stored for future reuse. This represents an important aspect of a circular of asset management, since functioning appliances can be stored until they are needed in another apartment or property. Without some form of storage solution, reuse practices would become considerably more difficult in practice, as appliances would either need to be immediately redeployed or risk being discarded entirely. However, the analysis also suggests that storage alone is insufficient. If the organization lacks a clear overview of which appliances are stored,

their condition, and their remaining lifecycle, the reuse hub itself risks becoming overstocked. inefficient and difficult to manage. (Stefan Hvalgren, Fixkod). Over time, this may result in fully functional appliances still being discarded due to insufficient storage capacity or limited visibility regarding available assets. Stefan Hvalgren (Fixkod) additionally emphasized that storage itself generates operational costs, particularly when appliances remain unused for extended periods of time.

These findings highlight the importance of visibility and traceability over both apartments and storage facilities for circular asset management practices to function effectively. This can be connected to Razak et al. (2023), who describe how traceability contributes to the earlier identification of disruptions and supports more proactive operational actions. In this context, improved traceability could help organizations identify when storage facilities are approaching capacity, which appliances risk becoming outdated, or which apartments are approaching future replacement needs. The findings can also be related to Alejandre et al. (2022), who argue that sustainability research related to appliances focuses primarily on energy efficiency rather than adopting a broader lifecycle perspective. This may contribute to situations where fully functional appliances are replaced, while the organization mainly collects information related to energy consumption rather than product history, usage patterns, maintenance history and remaining lifecycle. From this perspective, the lack of lifecycle-related information identified in the empirical findings becomes more understandable. Similar arguments are presented by Diop et al. (2021), who emphasize that decisions and information collection during the early stages of the lifecycle influence the long-term operability, maintenance efficiency, and sustainability outcomes. Without this information, organizations risk developing inefficient maintenance strategies and weaker long-term resource utilization practices. These findings can also be connected to Tsay et al. (2022), who demonstrate that information management becomes both costly and inefficient when data is fragmented across multiple systems, documents, and communication channels. At the same time, Delemere & Liston (2024) show that the willingness to repair rather than replace appliances already exists among both organizations and users. The challenge, therefore, does not necessarily stem from a lack of motivation, but rather from insufficient information visibility and the absence of systems capable of supporting these decisions effectively in practice.

Finally, the findings demonstrate that planning and forecasting are essential for avoiding disruptions within larger operational projects. As Stefan Hvalberg (Fixkod) described, uncertainty regarding future appliance needs may lead organizations to either procure too few or too many appliances, thereby increasing the risk of delays, higher costs, and inefficient resource utilization. This further emphasizes the need for improved traceability and information management in order to support more proactive and long-term lifecycle-oriented asset management. From this perspective, the prototype developed within this study can be understood as a potential support tool for reducing several of these barriers. By providing Operational Managers with improved visibility regarding appliance conditions, remaining lifecycle, repair status, and placement history, the system may improve both coordination and long-term management of reusable assets. The prototype also clearly indicates whether

apartments are currently utilizing reused appliances, which may further strengthen traceability and lifecycle management over time.

6.5 Organizational and Operational Constraints

Issues related to physical access have already been extensively discussed throughout this study and represent a central challenge within the empirical findings. Since many appliances are located inside tenants' homes, access to the assets becomes highly restricted, limiting both inspections and information gathering. However, beyond these physical limitations, the findings also indicate that several organizational and operational factors influence the possibility of implementing more traceable and lifecycle-oriented ways of working. As highlighted by Stefan Hvalgren (Fixkod), the real estate sector can be described as relatively conservative, where established routines and traditional ways of working remain dominant. According to the respondent, renting out apartments is not a new practice, and many organizations have therefore operated in similar ways for extended periods of time. Stefan Hvalgren (Fixkod) further connected this to issues of governance and organizational control, where organizations develop familiarity and security within existing structures, systems and workflows.

This can be connected to Seo (2013), who argues that resistance to new systems is often related to the perceived loss of control. Increased transparency and more standardized processes may reduce individual autonomy, particularly in environments where employees are accustomed to managing their own workflows independently. In addition, concerns regarding job security or changes in work roles may arise when systems automate tasks or alter established routines. Coşkun et al. (2022) further argue that such resistance can become especially strong within smaller organizations, where critical tacit knowledge is often concentrated around specific individuals who may perceive standardized systems as threats to their role and importance. This reluctance to share information may also be connected to the traditional view of information as a source of organizational power. However, Christopher & Lee (2004) emphasize that increased information sharing instead improves overall system performance by reducing uncertainty and enhancing organizational responsiveness.

Since the respondents in this study primarily represented larger organizations, the resistance may not necessarily stem from informal individual power structures to the same extent. Larger organizations are generally more accustomed to standardized systems and formalized operational processes, which may reduce some forms of resistance compared to smaller organizations (Coşkun et al., 2022; Haug et al., 2009). Despite this, the empirical findings still indicate organizational reluctance toward changing existing systems and routines. Respondents also described that organizations within the public sector often demonstrate slower adoption of new solutions due to perceived risks and limited incentives for change. Interestingly, this partially contrasts with Nicholson-Crotty et al., (2019), who argue that the perception of the public sector as inherently more risk-averse is more nuanced in practice. Their findings suggest that decision-making behavior within public organizations is highly

context-dependent and influenced by how risks and outcomes are framed. Consequently, resistance to change cannot solely be explained through individual risk aversion, but must be understood through organizational structures, institutional constraints, and contextual factors influencing decision-making processes. This may partly explain Stefan Hvalgren's (Fixkod) description of the real estate sector as conservative. From this perspective, resistance is not necessarily rooted in unwillingness to innovate, but rather in organizational structures and operational contexts that reinforce stability, continuity and predictability. This may also relate to organizational image and risk awareness within public-sector contexts, where maintaining organizational stability and avoiding perceived risks may influence attitudes toward change and innovation (Lee et al., 2023).

From an innovation perspective, Pedersen (2022) emphasizes the importance of individuals who initiate and drive organizational change. This can clearly be connected to the empirical findings, where Stefan Hvalgren (Fixkod) explained that successful implementation often requires an individual within the customer organization who believes in the innovation and acts as an internal ambassador. Such individuals may function as drivers of change by promoting adoption and creating internal support for new systems and processes. At the same time, previous research highlights that successful implementation also requires structured change management processes. According to De Búrca et al (2005), top management support, targeted training, and clear communication are central factors for increasing user acceptance and reducing uncertainty during implementation. This becomes particularly important in avoiding the emergence of "shadow systems" or inconsistent data handling, where employees continue relying on parallel solutions outside the official system (Haug et al., 2009; Seo, 2013). These tendencies were also visible within the empirical findings. Hampus Granström (Project Leader Internationalization) explained that one major obstacle is that organizations often continue solving problems using existing methods, even if those solutions may be less effective. He also emphasized that organizations may perceive new systems as obstacles if they cannot be integrated with already existing platforms and workflows.

Interestingly, similar concerns also appeared during the feedback interview regarding the prototype. The Operational Manager (Large Real Estate) pointed out that the prototype could become problematic if it operated as a separate system alongside the organization's existing platform. Managing multiple systems simultaneously was perceived as inefficient and risked increasing fragmentation rather than reducing it. This concern becomes understandable when connected to Tsay et al. (2022), who argue that fragmented systems and separated information structures reduce efficiency and limit organizations' ability to make informed and timely decisions. At the same time, the findings also indicate a broader challenge related to organizational dependency on established systems and routines. Transitioning from one system to another may involve significant operational effort, migration of large amounts of historical information, and extensive organizational adaptation processes. This remains true even when the existing systems are perceived as lacking important functionalities required for effective lifecycle and traceability management. Therefore, the findings further reinforce the importance of internal ambassadors, top management support, targeted training, and clear

communication for successful organizational implementation and adoption (Pedersen., 2022; De Búrca et al., 2005).

6.6 From Fragmentation to Structured Traceability

Building upon the previous sections, it becomes clear that fragmented and inaccessible information creates significant operational challenges if not properly addressed. As described by Razak et al. (2023), traceability refers to the ability to track and trace the history, location and status of products throughout their lifecycle. Within supply chains, traceability contributes to increased transparency and improved monitoring of operational processes. According to Razak et al. (2023), it further supports improved risk awareness, increased reliability, and enhanced trust between stakeholders. In addition, traceability enables earlier detection of disruptions and allows organizations to respond more proactively to operational issues. By supporting real-time tracking and information exchange, traceability may also strengthen organizational resilience and flexibility.

Despite these advantages, Razak et al. (2023) emphasize that traceability is still often limited in practice, particularly in environments where assets are physically distributed and difficult to access. This is highly relevant to the empirical findings of this study, where information was repeatedly described as fragmented, missing, or difficult to retrieve. For example, respondents explained that product history and lifecycle-related information was often difficult to locate in practice. Although the organizations already utilized centralized systems containing information connected to apartments and appliances, respondents emphasized that these systems lacked sufficient functionality regarding lifecycle history, accessibility of information, and long-term traceability. The Operational Manager (Large Real Estate) explained that, without resource or budget limitations, the ideal solution would involve a system where all information connected to apartments and appliances would be directly accessible. This would include installation dates, warranties, service history, maintenance records and other product related information. This suggests that a gap currently exists between the information required operationally and the level of accessibility provided by existing systems.

Similarly, Stefan Hvalgren (Fixkod) highlighted that many organizations already possess parts of the necessary information, but what is primarily missing is an efficient way of retrieving the correct information at the exact moment it is needed. This further supports the statements from the Operational Manager (Large Real Estate), who explained that though information exists within the current system, accessing it in practice is often both difficult and time-consuming. The fragmentation of information also became visible during appliance replacement processes. Respondents explained that identifiers are sometimes removed or not updated correctly during replacement processes when appliances are exchanged, resulting in important historical data and traceability being lost from the systems entirely. This creates uncertainty and limits the organization’s ability to maintain reliable lifecycle tracking over time.

Since resource efficiency was identified as a major concern in the empirical findings, improved traceability may also contribute to more effective resource utilization and cost reductions. Improved visibility of asset conditions, locations, and histories may reduce situations where technicians spend time repairing appliances that should instead be replaced, or where functioning appliances are discarded simply because relevant information is unavailable. The findings clearly indicate that organizations are interested in improving their current ways of working, but that they often lack practical methods capable of supporting such improvements. From this perspective, the QR-based traceability system developed within the prototype can be understood as a potential approach for addressing several of these challenges.

Within the prototype, each appliance is assigned a unique QR-based identifier at the point of registration. This identifier then follows the appliance throughout its lifecycle and connects directly to all related information, including installation date, warranty information, service history, reports, repair logs, and location. The QR codes are integrated into several operational workflows, including fault reporting, technician processes, maintenance tracking, reuse activities and lifecycle management. Since updates are continuously registered directly within the system, the prototype supports more structured information management and improved historical traceability over time. At the same time, the findings also indicate that successful implementation would depend on organizations' consistency in maintaining and updating asset identifiers correctly. For example, newly installed appliances would need to be properly registered and connected to the correct apartment within the system. However, the prototype suggests that such processes could be significantly easier through standardized QR-based workflows and centralized information management.

The findings additionally indicate that implementation success depends not only on technological functionality, but also on user acceptance and organizational adoption. Employees must be willing and prepared to use the system consistently to avoid issues such as incomplete registrations, inconsistent data handling, or the emergence of shadow systems might appear (De Búrca et al. 2005). At the same time, the prototype does not fundamentally aim to change existing operational workflows. Instead, it seeks to simplify current work practices by making information easier to access, update, and track throughout the appliance lifecycle.

From this perspective, improved traceability may contribute not only to increased operational visibility but also to reduced uncertainty, improved resource efficiency, lower operational costs, and more sustainable lifecycle-oriented asset management practices.

6.7 Enabling Data-Driven and Sustainable Decision-Making

Combining the findings from the previous sections, it becomes evident that the identified challenges are closely interconnected across both the empirical findings and the theoretical framework. The analysis demonstrates that fragmented information, limited visibility,

uncertainty, and organizational barriers negatively affect lifecycle management, operational decision-making, and long-term resource utilization. At the same time, the findings also indicate several potential ways of addressing these challenges. From this perspective, the developed prototype becomes particularly relevant, as it was specifically designed to support many of the identified operational and informational challenges. Through the QR-based tracking system, the prototype enables improved control over appliances and simplifies the tracking of information throughout the entire appliance lifecycle. By connecting all relevant information directly to individual appliances, the system supports more structured and continuous information management. This includes lifecycle status, warranty information, repair history, installation dates, location, forecasting data, and maintenance-related records. The prototype can therefore also be understood as a decision-support system. By centralizing lifecycle-related information and improving traceability, the system may support Operational Managers in making more informed decisions regarding repair, replacement, reuse, procurement, and long-term planning activities. Rather than relying on assumptions or incomplete information, decision-making processes may become more structured, proactive, and data-driven.

The findings throughout this study strongly indicate that high-quality data is essential for effective decision-making. As emphasized by Berisha-Shaqiri (2014), access to accurate, relevant, and timely information improves decision quality, reduces inefficiencies, and strengthens operational performance. Similarly, Haug et al. (2009) and Seo (2013) highlight how inconsistent or fragmented information reduces system effectiveness and increases the risk of errors and inefficiencies. By enabling more structured and centralized data management, the prototype aims to reduce fragmentation while improving consistency and accessibility. For example, through the home appliance interface (see Figure 5), users can quickly access apartment-specific information, including warranty status, lifecycle indicators, serial numbers, appliance type, and service history. This directly addresses the difficulties highlighted in the empirical finding, where respondents described access to appliance-related information as highly time-consuming and, in some cases, nearly impossible (Stefan Hvalberg, Fixkod).

An important aspect of the prototype is that all functionalities are interconnected. Information entered within one part of the system automatically updates corresponding sections throughout the platform. This creates a more consistent information flow while simultaneously reducing the risk of fragmented communication between departments and operational processes. Such integration also improves coordination between tenants, technicians, Operational Managers, forecasting functionalities, and sustainability reporting. The forecasting functionality becomes particularly relevant from a decision-making perspective. By visualizing future costs, expected replacement needs, and maintenance expenses, the system supports more proactive planning and resource allocation. This directly addresses the uncertainty discussed by Schultz et al. (2010), who explain that uncertainty increases when outcomes become difficult to predict and when decisions involve risk or potential losses. Within the empirical findings, respondents described how uncertainty often leads organizations to “round up” procurement volumes or prematurely replace appliances in

order to minimize future operational risks. By improving access to structured information and future forecasting, the prototype may reduce uncertainty and support more data-driven decision-making. This may, for example, reduce the risk of purchasing incorrect quantities of appliances, avoid delays caused by missing replacements, and decrease the need for time-consuming information gathering. Through improved visibility of future needs and lifecycle status, organizations may transition from reactive decision-making towards a more proactive lifecycle-oriented planning processes.

The reuse hub further supports this more proactive and lifecycle-oriented way of working (see Figure 20). Through the reuse hub, managers gain an overview of reusable appliances, including lifecycle status, condition, warranty information, repair needs, storage location, and readiness for reuse. Appliances can then be directly assigned to technicians, who receive instructions regarding pickup, transportation, installation, and status updates. Once installation is completed, the information is automatically updated throughout the system, thereby ensuring continuous traceability and reducing manual administrative work.
This centralized, integrated way of working directly addresses the challenges identified by Tsay et al. (2022), who describe fragmented information systems and disconnected data management as both costly and inefficient within facility management contexts. Similarly, Berisha-Shaqiri (2014) argues that poorly structured information systems limit organizations' ability to effectively utilize data, particularly within asset-intensive environments.
The prototype additionally reduces some of the risks associated with manual data management. Tsay et al. (2022) note that manually entered information increases the risk of inaccurate data and human error. Although appliances still require initial registration within the system, the prototype attempts to simplify this process through standardized QR-based registration and structured workflows. By making registration processes more accessible and intuitive, the system aims to reduce resistance while simultaneously improving data quality and consistency.

From a sustainability perspective, the prototype also addresses challenges discussed by Alejandre et al. (2022), who argue that organizations often focus primarily on energy efficiency while neglecting a full lifecycle perspective. This may contribute to premature replacements of functioning appliances. By instead emphasizing lifecycle tracking, repair history and reuse opportunities, the prototype supports more sustainable long-term resource utilization practices. This perspective also aligns with Diop et al. (2021), who emphasize that decisions and information established during the early lifecycle stages significantly influence long-term operability, maintainability, and sustainability outcomes. Through continuous lifecycle tracking and historical traceability, the prototype may therefore support more informed decisions regarding repair, replacement, and reuse activities. Another important aspect of the prototype is the sustainability dashboard. All activities related to repair, reuse, and lifecycle extension are automatically visualized within the system. This allows organizations to monitor both environmental impact and financial savings over time. The dashboard is intentionally designed not only as a reporting functionality, but also as a motivational and communicative tool that intended to highlight the positive outcomes associated with sustainable practices. The sustainability visualization is presented through two

complementary approaches. On the home dashboard, environmental impact is visualized in simplified and more intuitive indicators, such as trees saved or environmental impact (see Figure 1). This allows users to quickly understand the positive impact of their work without needing to interpret extensive numerical data. Simultaneously, a more advanced sustainability interface provides detailed information regarding environmental savings, reuse levels, lifecycle extension, and sustainability performance across different appliance categories. This functionality may also contribute to increased organizational motivation and acceptance of more sustainable workflows by making both environmental and financial benefits more visible. Such visibility may further strengthen organizational willingness to allocate resources towards reuse processes, traceability systems, and lifecycle-oriented asset management initiatives.

At the same time, the findings indicate that technology alone is insufficient. As emphasized by Christopher & Lee (2004), effective coordination depends on strong information sharing and visibility across organizational processes in order to reduce uncertainty and improve responsiveness. Similarly, De Búrca et al. (2005) emphasize the importance of structured change management, including top management support, targeted training, and clear communication in order to increase use acceptance and reduce resistance toward new systems. Without organizational support and proper implementation strategies, organizations risk continued fragmentation, inconsistent use, and the emergence of shadow systems, as highlighted by Haug et al. (2009) and Seo (2013). Therefore, although the prototype addresses many of the operational and informational challenges identified in this study, its effectiveness would still depend on organizational willingness to adopt, maintain and utilize the system in practice.

Overall, the analysis suggests that improved traceability, centralized information management, and lifecycle-focused systems may support more proactive, sustainable, and data-driven decision-making within property management contexts.

7. Conclusions

The purpose of this study was to analyze challenges related to information management and traceability of household appliances within property management, and to explore how a QR-based traceability system could support improved visibility, decision-making, and sustainable resource utilization. The following section presents the main findings of the study through answers to the research questions.

RQ1: How is information about household appliances currently managed in property management, and what challenges arise due to limited traceability and information accessibility?

The findings show that information regarding household appliances is currently managed through centralized property management systems, where appliance-related information is registered together with other operational and apartment-related data. When appliances are

replaced or maintained, the information is updated within the system. Fault reports are generally initiated by tenants, after which technicians are sent to assess the problem before reporting their evaluation back to management, who subsequently make decisions regarding repair or replacement. The findings indicate that these decisions are often strongly influenced by cost considerations and practical constraints. For example, appliances that are considered old, such as products approximately eight to ten years old, are often replaced rather than repaired. At the same time, the empirical findings show that important lifecycle-related information is frequently fragmented, difficult to access or inconsistently registered across systems and workflows.

A central challenge identified in the study is that household appliances are physically distributed within tenants' homes, making access to assets difficult. This creates operational barriers for inspections, lifecycle tracking, and information collection. Although the organizations already utilize centralized systems, respondents emphasized that existing systems are not perceived as sufficient for supporting long-term lifecycle management and traceability. As a result, organizations often rely on experience and simplified assessments rather than structured lifecycle data. Respondents expressed a need for systems capable of providing clearer apartment-level visibility, where all appliances connected to a specific apartment can easily be identified together with their lifecycle status, service history, warranty information, and location.

The findings further demonstrate that fragmented information and limited visibility negatively affect an organization's ability to fully utilize appliance lifecycles. Missing or inaccessible information contributes to uncertainty in decision-making processes, resulting in more reactive and risk-minimizing decisions, where replacement is often perceived as the safer and more efficient alternative compared to repair or reuse. Limited visibility also affects forecasting and long-term planning, making it difficult for organizations to anticipate future replacement needs and allocate resources proactively. Consequently, appliances may be prematurely replaced despite still having remaining functional value.

RQ2: How can a QR-based system be designed to improve information management and visibility?

The findings suggest that a QR-based system can improve information management and visibility by assigning unique digital identifiers to individual appliances throughout their lifecycle. Through QR-based identification, appliances become easier to track, update, and manage across different operational processes within property management. The developed prototype demonstrates how all functionalities within the system can be interconnected through these identifiers, allowing information to synchronize across different parts of the system automatically. This creates a more integrated and centralized information flow, where relevant stakeholders can access up-to-date lifecycle-related information in real time.

Within the system, QR-codes support integrated operational workflows by allowing technicians to verify appliances and update maintenance information directly within the system, while tenants can report issues connected to specific appliances through QR-based

reporting functionalities. The information is automatically synchronized and becomes accessible to Operational Managers and other relevant actors in real time. This improves communication, reduces fragmented information handling, and strengthens lifecycle visibility across different stakeholders and operational processes.

The QR-based structure also supports reuse and lifecycle-oriented asset management practices. Throughout the reuse hub, organizations can track which appliances are available, reserved, installed, undergoing refurbishment, or approaching end-of-life. This improves visibility of reusable assets while supporting coordination, forecasting, and long-term lifecycle planning. Overall, the findings indicate that QR-based traceability systems may support more transparent and accessible information management by centralizing lifecycle-related information and improving visibility across operational processes. This may contribute to improved forecasting, easier reporting, increased traceability, and more data-driven decision-making regarding repair, reuse, replacement, and resource allocation. In addition, the visualization of financial savings and sustainability outcomes within the system may further support organizational motivation and strengthen willingness to adopt more lifecycle-oriented and resource-efficient ways of working.

RQ3: How can such a system support decision-making and contribute to more sustainable resource utilization?

The findings indicate that a QR-based traceability system can support decision-making and contribute to more sustainable resource utilization by improving visibility, accessibility, and coordination of lifecycle-related information. By centralizing appliance-related data within one integrated system, organizations gain easier access to the information required for making informed operational and strategic decisions. The system supports decision-making by providing relevant lifecycle information directly connected to individual appliances, including maintenance history, warranty status, condition, repair records, replacement history, forecasting data, and reuse status. This reduces the need for decisions to be based on assumptions or incomplete information and instead enables more structured and data-driven decision-making processes.

Improved visibility also contributes to more proactive planning and forecasting. Through forecasting functionalities and lifecycle indicators, organizations can better anticipate future replacement needs, maintenance activities, procurement volumes, and resource allocation requirements. In addition, historical lifecycle data allows organizations to evaluate previous decisions and operational outcomes, thereby supporting continuous learning and improved long-term planning. From a sustainability perspective, the system contributes to more resource-efficient lifecycle management by improving an organization's ability to repair, reuse, and extend the lifespan of existing appliances. By increasing visibility regarding appliance condition, remaining lifecycle, and reuse availability, organizations may reduce unnecessary replacement of functional appliances and instead make more sustainable lifecycle-oriented decisions.

The reuse hub functionality further strengthens this process by integrating reusable appliances directly into operational workflows rather than treating reuse as a separate or secondary

activity. Through centralized visibility of reusable assets, organizations can more effectively coordinate refurbishment, relocation, installation, and lifecycle tracking of appliances across different properties. In addition, the visualization of financial savings and sustainability outcomes may strengthen organizational motivation and increase willingness to continue working with lifecycle-oriented and circular resource management practices. By clearly demonstrating both environmental and economic benefits, the system may contribute to increased acceptance of sustainable operational strategies and support long-term organizational commitment toward more resource-efficient ways of working. Overall, the findings suggest that a centralized traceability system may support more proactive, transparent and sustainability-oriented decision-making within property management contexts.

7.1 Prototype Contribution

Based on the findings in this study, the developed prototype can be understood as a proof-of-concept demonstrating how several of the identified operational and informational challenges may be addressed in practice. The prototype was specifically designed around the empirical findings, and theoretical concepts identified throughout the study, particularly regarding fragmented information, limited lifecycle visibility, and insufficient traceability of household appliances. The prototype illustrates how a QR-based traceability system may support more structured and lifecycle-oriented asset management by centralizing appliance-related information within one integrated system. Through functionalities connected to lifecycle tracking, warranty management, maintenance history, forecasting, sustainability reporting, and reuse coordination, the prototype demonstrates how improved visibility and traceability may support informed and proactive decision-making processes. In particular, the prototype addresses several challenges identified within the empirical findings, such as difficulties related to tracking warranties, maintenance history, appliance condition, and lifecycle status over time. By connecting all lifecycle-related information directly to individual appliances through QR-based identifiers, the system demonstrates how information can become more accessible, transparent, and easier to manage throughout the asset lifecycle.

The prototype also illustrates how multiple stakeholders, including tenants, technicians, and Operational Managers, can be integrated into the same information flow. This contributes to improved coordination, reduced fragmentation, and more transparent operational processes. In addition, the integrated reuse hub, forecasting functionalities, and sustainability indicators demonstrate how digital traceability systems may support both operational efficiency and more sustainable resource utilization practices. Overall, the prototype demonstrates the practical feasibility of applying QR-based traceability and lifecycle-oriented management principles within property management contexts. Although the prototype remains exploratory and not fully implemented, it illustrates how digital traceability systems may bridge the gap between theoretical concepts related to visibility, traceability, sustainability, and practical operational workflows within property management.

7.2 Overall Conclusion

To conclude, the findings of this study demonstrate that fragmented information, limited visibility, and insufficient traceability create significant operational challenges within property management contexts. These challenges contribute to uncertainty, reactive decision-making, inefficient utilization, and difficulties in fully utilizing the lifecycle of household appliances. The study further indicates that QR-based traceability systems may support improved visibility and lifecycle-oriented asset management by centralizing appliance-related information and connecting operational workflows through shared digital identifiers. By improving accessibility of lifecycle-related information, such systems may support more proactive, transparent, and data-driven decision-making processes regarding repair, reuse, replacement, forecasting, and resource allocation. The findings also highlight that operational efficiency and sustainability are closely interconnected. Through improved lifecycle tracking, reuse coordination, forecasting, and information visibility, organizations may reduce unnecessary replacement of functional appliances while simultaneously improving financial efficiency and reducing environmental impact. From this perspective, lifecycle-oriented asset management may contribute both to operational improvements and more sustainable resource utilization practices.

At the same time, the study demonstrates that technology alone is insufficient for creating successful organizational change. The effectiveness of such systems depends heavily on organizational adoption, integration with existing workflows and systems, consistent information management practices, and user acceptance. Without organizational support and structured implementation processes, the potential benefits of improved traceability and lifecycle management may be difficult to fully realize in practice. Overall, the study suggests that QR-based traceability systems may function as a valuable support tool for improving visibility, decision-making, and sustainability within property management, while also demonstrating the importance of integrating technological solutions with organizational and operational practices.

8. Contributions

This study contributes both practical and theoretical contributions within the fields of SCM, information systems, lifecycle-oriented asset management, and sustainability within property management contexts.

From a practical perspective, the study provides insights into current operational challenges related to the fragmented information, limited traceability, and restricted visibility of household appliances within property management operations. The findings illustrate how difficulties in accessing lifecycle-related information affect maintenance processes, forecasting, reuse activities, and decision-making regarding repair and replacement. In addition, the study demonstrates how a QR-based traceability system may support more structured and centralized information management. Through the development and evaluation of the prototype, the study illustrates how digital traceability solutions can improve accessibility of asset-related information, strengthen lifecycle visibility, and support more informed and data-driven decision-making processes. The prototype further demonstrates how integrated information flows may contribute to improved coordination between tenants, technicians, Operational Managers, sustainability reporting, and forecasting activities. The study also contributes practical insights regarding how lifecycle tracking, reuse management, forecasting functionalities, and sustainability visualization may support more resource-efficient and proactive asset management practices. In this way, the proposed system illustrates how organizations may reduce unnecessary replacement of functional appliances, improve resource utilization, and support more sustainable operational practices over time.

From a theoretical perspective, the study contributes to existing research on internal SCM by illustrating how concepts related to visibility, traceability, and information flow can be applied within property management environments. While previous SCM and traceability research has primarily focused on traditional industrial and inter-organizational supply chains, this study extends these concepts into internal property-management contexts involving distributed and physically inaccessible assets located within residential environments. This study further contributes to research on lifecycle-oriented asset management by highlighting how limited accessibility of lifecycle-related information affects operational decision-making, forecasting capabilities, and circular resource utilization practices. In addition, the findings contribute to sustainability and circular economy research by demonstrating how improved traceability and centralized information management may support repair, reuse, lifecycle extension, and reduced waste generation.

Overall, the study contributes to bridging the gap between SCM theory and practical application within property management by demonstrating how digital traceability systems may support more proactive, sustainable, and data-driven lifecycle management practices.

9. Limitations

This study is subject to limitations that should be considered when interpreting the findings.

One limitation is the relatively short timeframe of approximately ten weeks, which restricted the amount of empirical data that could be collected and analyzed. Although the available time was utilized efficiently, a longer research period could have enabled broader data collection, additional interviews, and more extensive prototype development and evaluation. Another limitation concerns the sample size and organizational scope of the interviews. The study primarily focused on perspectives from a larger real estate organization, which may limit the transferability of the findings to smaller organizations operating under different conditions and resource constraints. Future studies would therefore benefit from including a broader range of organizations, including both smaller and larger companies within both the private and public sectors. The study could also have benefited from collecting feedback from a wider range of employees and organizational roles regarding workflows, operational routines, and information management practices. Including additional perspectives may have provided a more comprehensive understanding of organizational challenges and implementation conditions. In addition, since more interviews were conducted within a public-sector-related context, there is a possibility that respondents moderated or withheld certain opinions or problems due to concerns regarding organizational reputation or external perceptions, which can be related to Lee et al. (2023), who emphasize that public organizations often face higher levels of reputational pressure. Although the study followed ethical guidelines, ensured anonymity, and clearly communicated how the information would be used, it cannot be fully excluded that respondents were cautious in some of their responses.

A limitation concerning the QR-based solution itself is that although QR codes can support traceability by connecting physical appliances to digital information, their effectiveness depends on correct placement, readability, and consistent use. If a QR label is damaged, removed, incorrectly placed or not scanned by users, the connection between the physical appliance and the digital record may be weakened. In addition, the system is dependent on the quality of the data entered. Incorrect, outdated or incomplete information could lead to misleading lifecycle assessments and reduce the readability of decision-making. Therefore, QR-based traceability should not be viewed as a complete solution on its own, but as a part of a broader information management process that requires routines, data quality control, and organizational commitment.

Another limitation is that the prototype developed within this study was exploratory and not fully implemented within a real operational environment. Consequently, the feedback collected regarding the system was based on perceived usefulness and anticipated workflows rather than long-term practical usage. Future implementation studies would therefore be necessary to evaluate how the system performs in practice and how users interact with it over time. Finally, the study is primarily focused on household appliances within property management contexts. Therefore, the findings may not be directly transferable to other asset categories or industries without further adaptation and investigation.

10. Future Research

Future research could focus on evaluating the prototype through real-world implementation studies within a property management organization. Since the prototype developed in this study was exploratory and not fully functional, further research is needed to investigate how such a system would perform in practice and how users would interact with it during daily operational work. In addition, future studies could investigate user adoption and organizational acceptance of QR-based traceability systems within property management contexts. This includes examining how technicians, Operational Managers, and tenants interact with the system, as well as identifying factors that influence long-term adoption and consistent system usage.

Another important area for future research concerns system integration. Respondents emphasized the importance of integrating new traceability solutions with already existing property management systems in order to avoid fragmented workflows and multiple disconnected platforms. Future research could therefore investigate how QR-based traceability systems can be integrated within existing procurement management and enterprise systems, as well as what technical and organizational adaptations would be required for successful implementation. Further research could also investigate whether the organization would benefit from transitioning fully toward the more integrated lifecycle-oriented management system after further technological development and validation. Since the prototype developed in this study was not fully operational, future implementation studies would be necessary to evaluate its practical usability, effectiveness, and long-term organizational value.

Another interesting direction for future research concerning predictive maintenance and AI. Through the collection of historical lifecycle and maintenance data, future systems may partially support AI-driven forecasting and predictive maintenance functionalities capable of identifying failure risks before problems occur. This could further improve operational planning, maintenance efficiency, and resource utilization. Finally, future research could broaden the scope beyond household appliances and investigate whether similar traceability and lifecycle-management approaches could be applied to additional asset categories within property management. Examples may include elevators, doors, ventilation systems, or other distributed building-related assets. This could contribute to a broader understanding of how traceability systems may support lifecycle-oriented management practices across the real estate sector as a whole.

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Appendices

Appendix A

Interview Guide – Operational Manager

Can you describe your role and responsibilities related to property management or asset management?

How are household appliances currently managed within your organization?

How is information regarding household appliances stored and used today (e.g., age, condition, maintenance history)?

What challenges do you experience when it comes to accessing or using this information?

How are decisions currently made regarding whether an appliance should be repaired or replaced?

To what extent do you have visibility regarding where specific appliances are located?

Can you provide examples of situations where lack of information has affected decision-making?

Do you experience that appliances are sometimes replaced unnecessarily? If so, why?

How are renovation projects or appliance replacement processes usually handled?

How do you currently track appliances between different properties or apartments?

How does limited access to tenants' homes affect your ability to assess the condition of appliances?

What type of information would you need in order to make better decisions?

What would an ideal system for managing household appliances look like?

Appendix B

Interview Guide – Fixkod

Can you briefly describe your role and how Fixkod works with the real estate sector?

What type of customers do you primarily work with?

What types of problems have you seen that property owners experience regarding household appliances or other assets?

How did you first identify these problems?

- Are these recurring problems?
- Are customers themselves aware of these problems?

How do your customers currently manage information regarding household appliances?

What type of data is most often missing?

What do these shortcomings lead to in practice?

How do you view the challenge that assets are located inside tenants' homes?

How does this affect customers' ability to maintain control and visibility?

How do customers currently make decisions regarding repair or replacement of products?

To what extent are these decisions based on data versus experience?

Is there resistance toward new digital solutions?

What do customers request in a solution?

What is required for customers to use a system in practice?

How do customers discuss sustainability in relation to these issues?

Is sustainability perceived as a driving force or more as an external requirement?

How do you believe this area will develop in the future?

What is required for the industry to change its way of working?

Appendix C

Interview guide - Project Leader Internationalization

Can you describe your role and how you work?

How do you usually work to understand customer and user needs?

How can one determine whether a solution actually creates value for the user?

What is the most common mistake startups make when developing digital solutions?

What is required for a solution to actually be adopted and used in practice?

When does an idea or solution fail?

You mentioned that tools should be simple and create value, what does that mean in practice?

How important is it to minimize the user's workload within a system?

What happens if a system is perceived as too complex?

If we look at property management, what do you believe are the most important user needs?

What type of information needs to be quick and easy to access?

How important is it that the system fits into existing workflows and ways of working?

What is your spontaneous reaction to a QR-based solution for managing household appliances?

What determines whether such a solution will actually be used in practice?

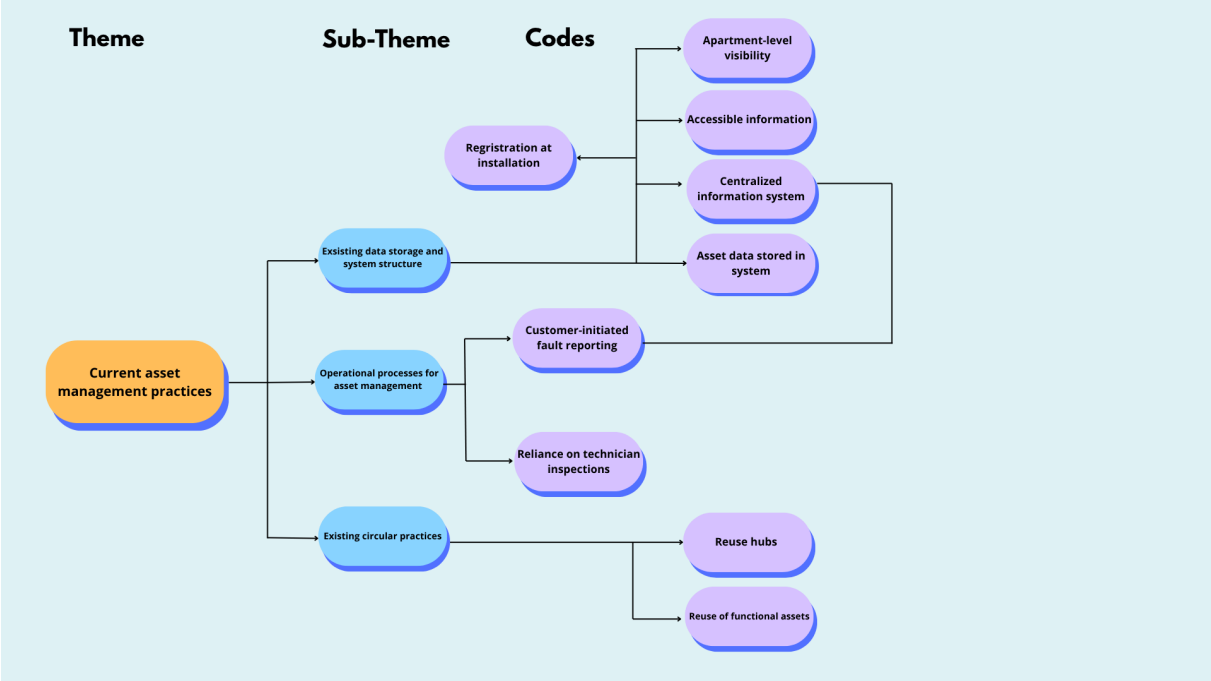
What could become barriers to implementation?

How could the data generated from such a system create value over time?

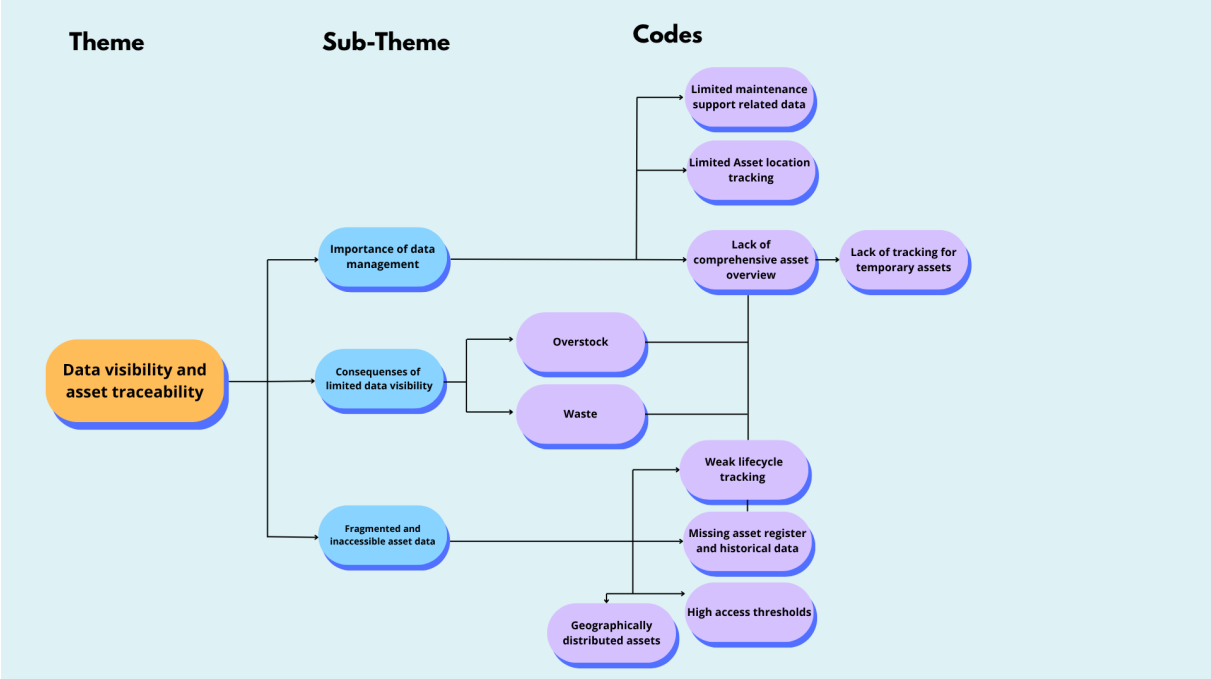
What could go wrong when implementing this type of solution?

Is there a risk that the system would not be used even if the solution itself is good?

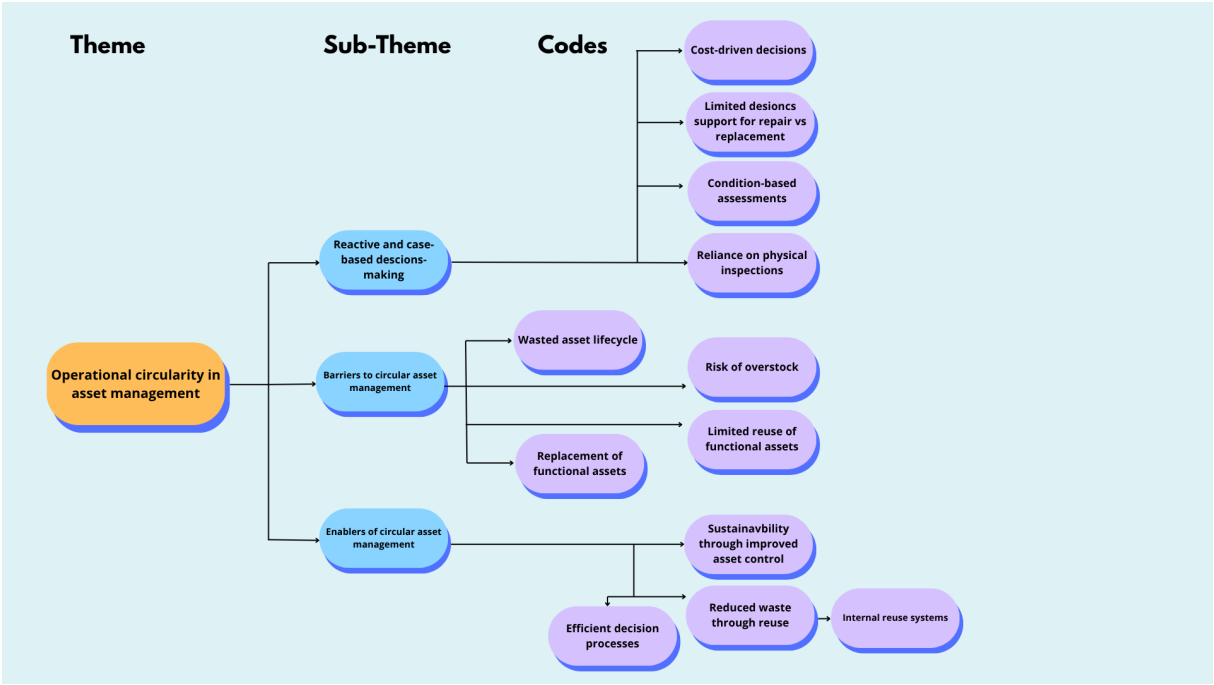
Appendix D



Appendix E



Appendix F



Appendix G

